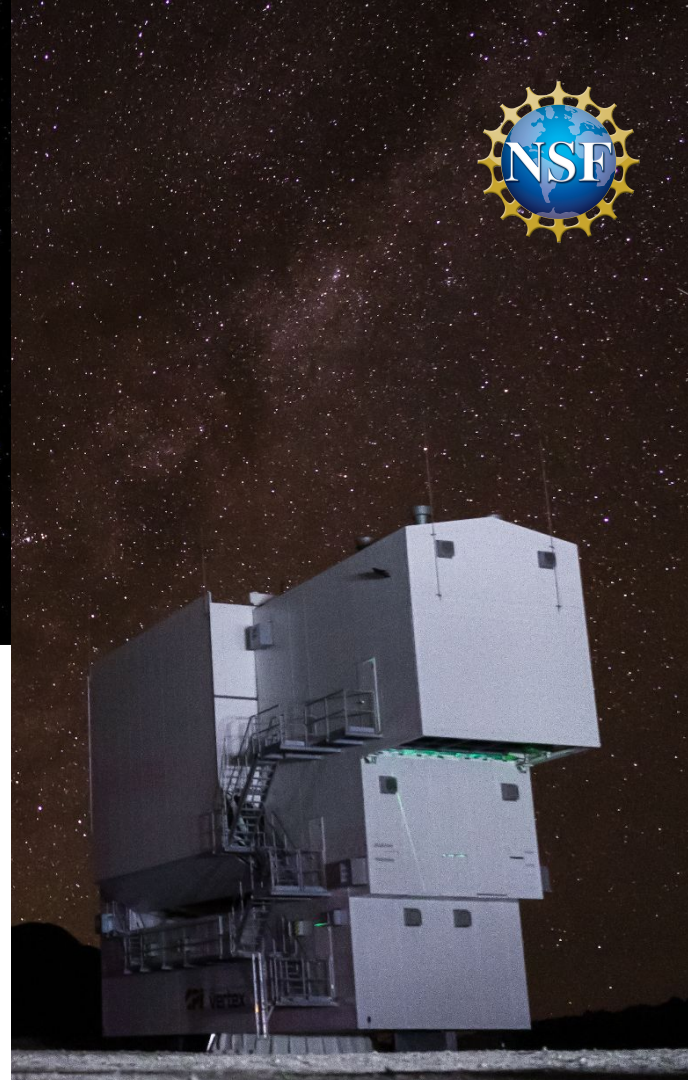


Working with CMB data



Mat Madhavacheril
University of Pennsylvania

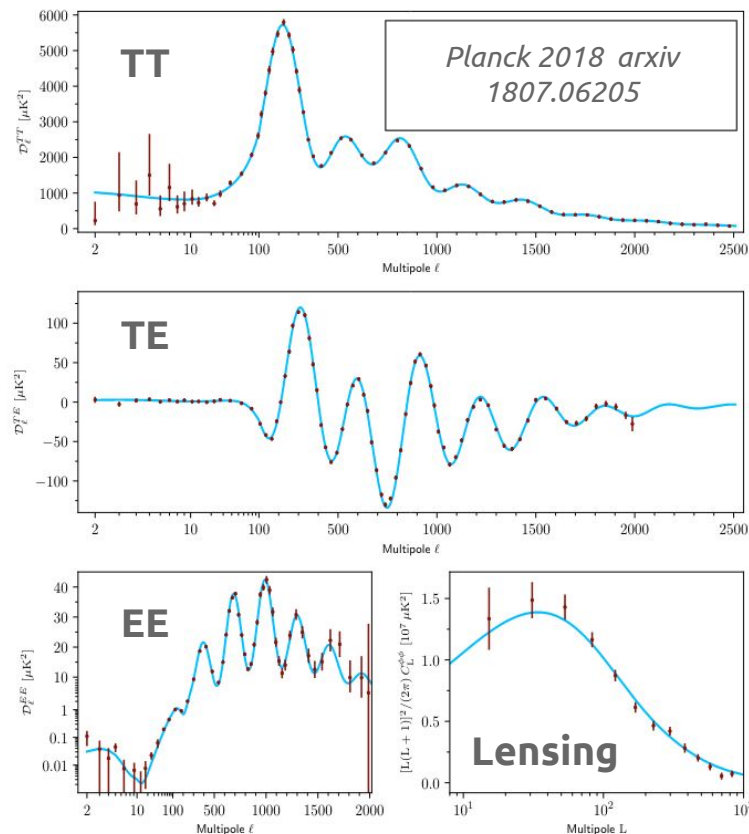
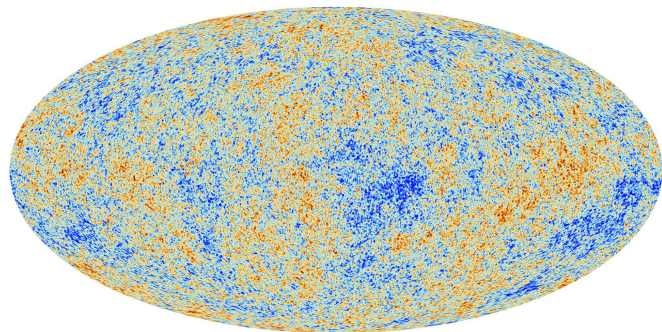


Outline

- ACT legacy; SO future
- Typical products produced by CMB experiments
- Where to find these products
- Conventions, units
- Noise
- Cyl. projection + rect. pix vs. Healpix
- What secondaries are in the maps
- Subtleties with stacking and ILC
- Polarization
- Links to resources

The WMAP+Planck Legacy -> next generation CMB

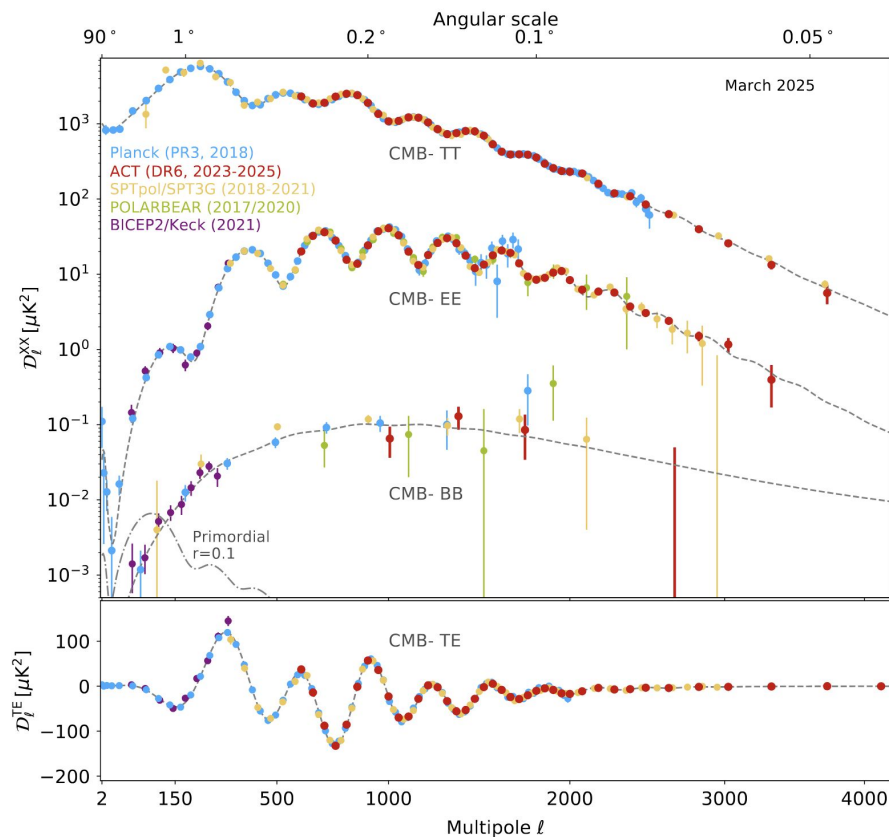
- Precision concordance Λ CDM model
- Init. Conditions: Gaussian, adiabatic, slightly red near scale-invariant power
- Dark matter needed to explain acoustic oscillations
- Acceleration due to dark energy (indep. evidence from CMB)
- Three relativistic/light neutrinos, $N_{\text{eff}} \sim 3$



The WMAP+Planck Legacy -> current generation CMB

Louis et al ACT 2025

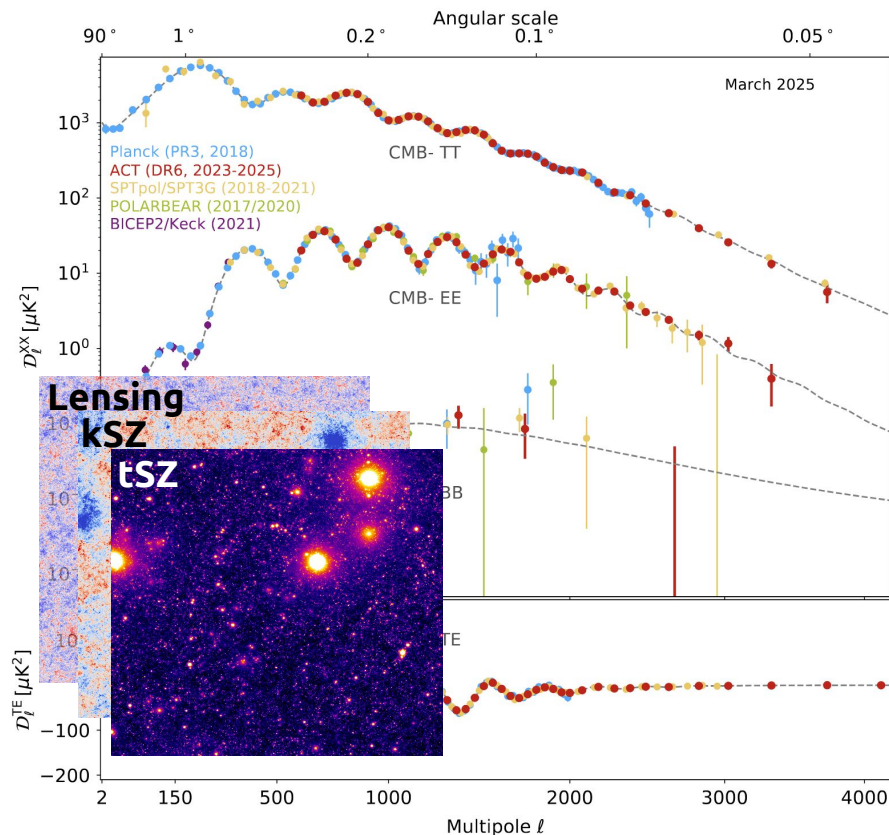
- Explore space of models for H_0 tension
- New light particles ΔN_{eff}
- Dark matter interactions
- Inflationary gravitational waves, r with B-modes



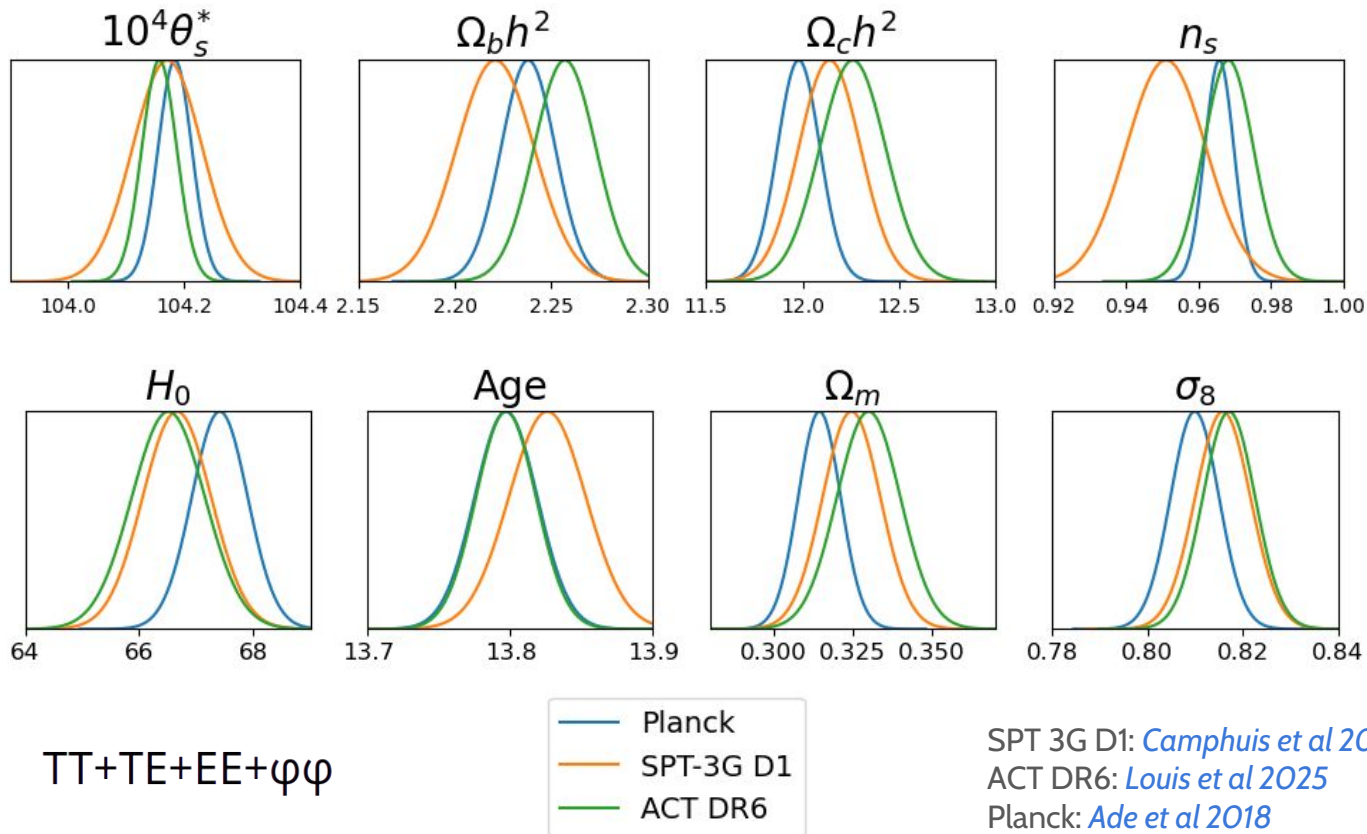
The WMAP+Planck Legacy -> current generation CMB

Louis et al ACT 2025

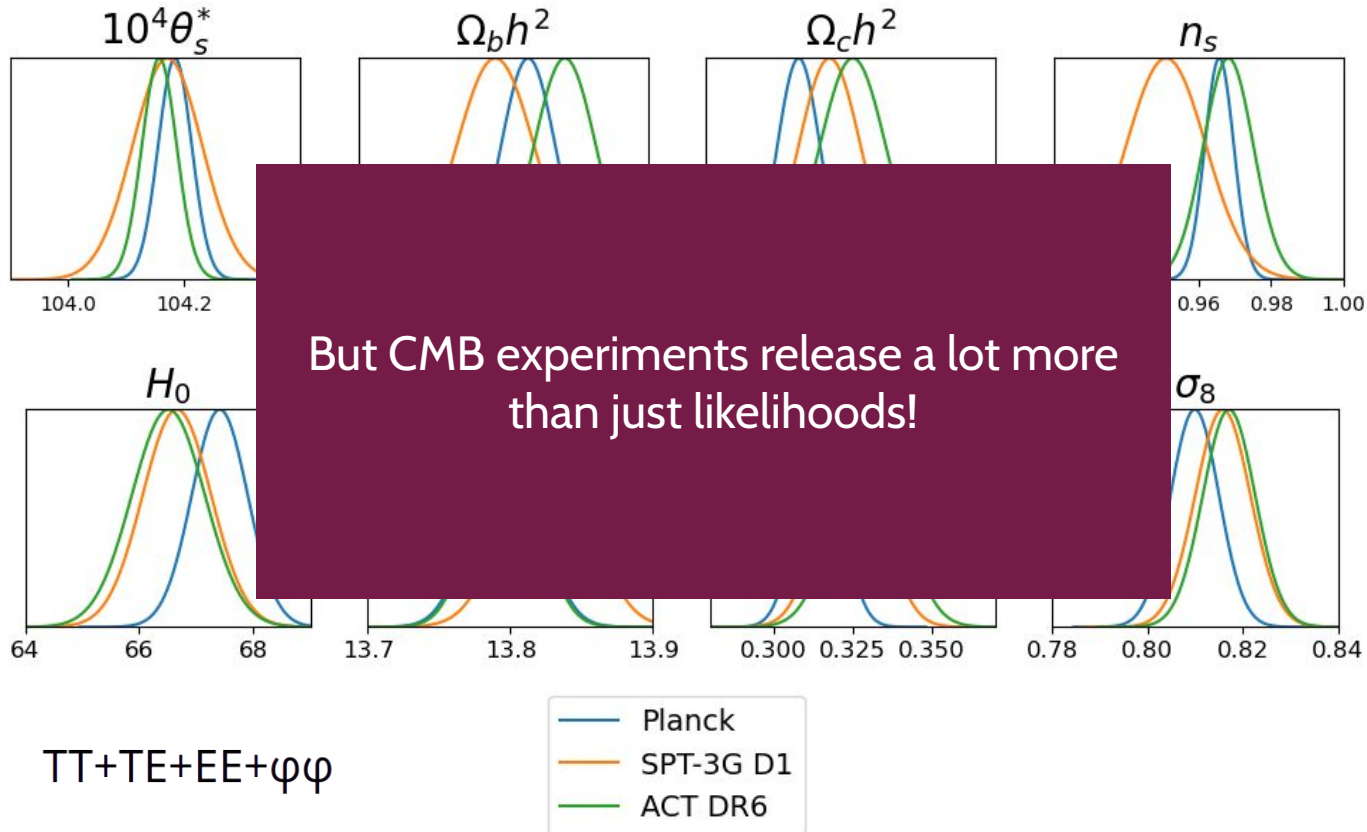
- Explore space of models for H_0 tension
- New light particles ΔN_{eff}
- Dark matter interactions
- Inflationary gravitational waves, r with B-modes
- Reconstructions and cross-correlations of mass (lensing), gas pressure (tSZ), velocities and gas density (kSZ)
 - Measure neutrino mass Σm_ν
 - Precision tests of non-Gaussianity from multi-field inflation f_{NL}
 - Measure baryonic feedback, constrain galaxy formation



Excellent agreement of LCDM parameters

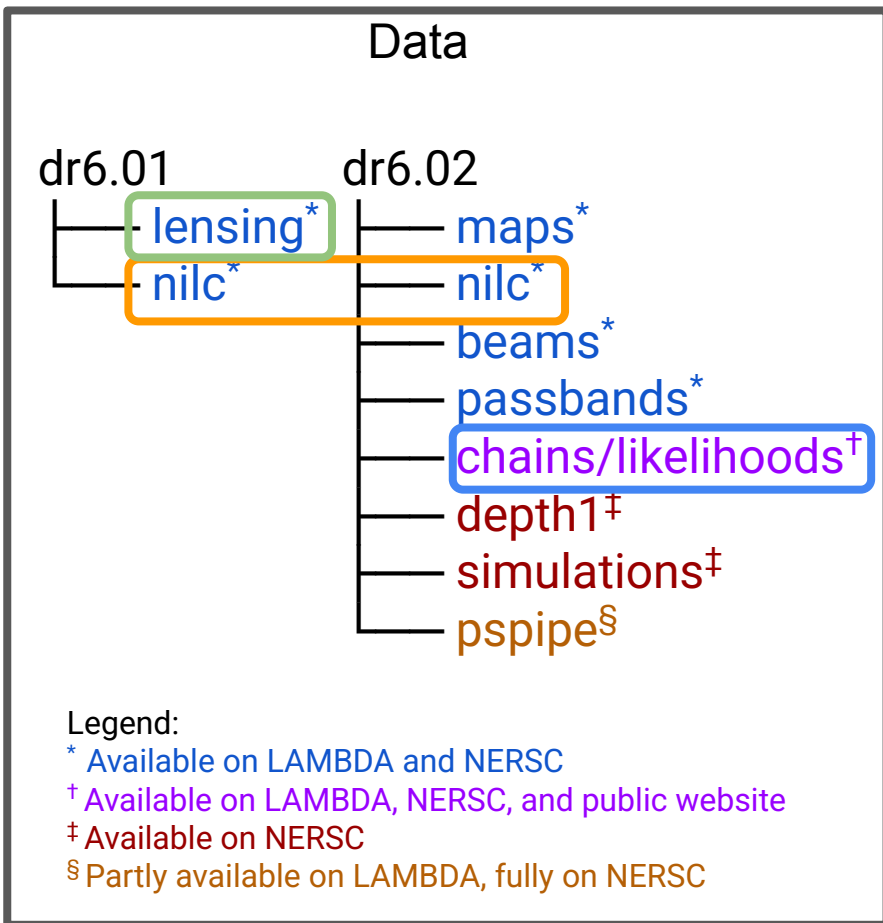


Excellent agreement of LCDM parameters



ACT DR6 Data Release – Quick Reference

Slide from Zach Atkins



Access



/global/cfs/cdirs/cmb/data/act_dr6/dr6.02
[Globus portal](#)

Resources



DR6_Notebooks Public

Python notebooks with DR6 tutorials:
github.com/ACTCollaboration/DR6_Notebooks

For questions reach out to
act_data_support@googlegroups.com or
act_notebooks@googlegroups.com

“CMB Units”

Dominant component at 90/150 GHz is CMB with Planck distribution. Calibrated experiment measures specific intensity (units = Jy/sr = $10^{-26} \text{ W}\cdot\text{m}^{-2}\cdot\text{Hz}^{-1}/\text{sr}$).

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}.$$

For small anisotropies

$$I_\nu \approx B_\nu(T_{\text{cmb}}) + \left. \frac{dB_\nu}{dT} \right|_{T_{\text{cmb}}} \Delta T.$$

So fluctuations in the intensity

$$\Delta I_\nu = g(\nu) \Delta T,$$

CMB maps are usually normalized into these “thermodynamic CMB temperature” units, thus independent of frequency. When working with bright sources, more meaningful to convert back to Jy.

“CMB Units”

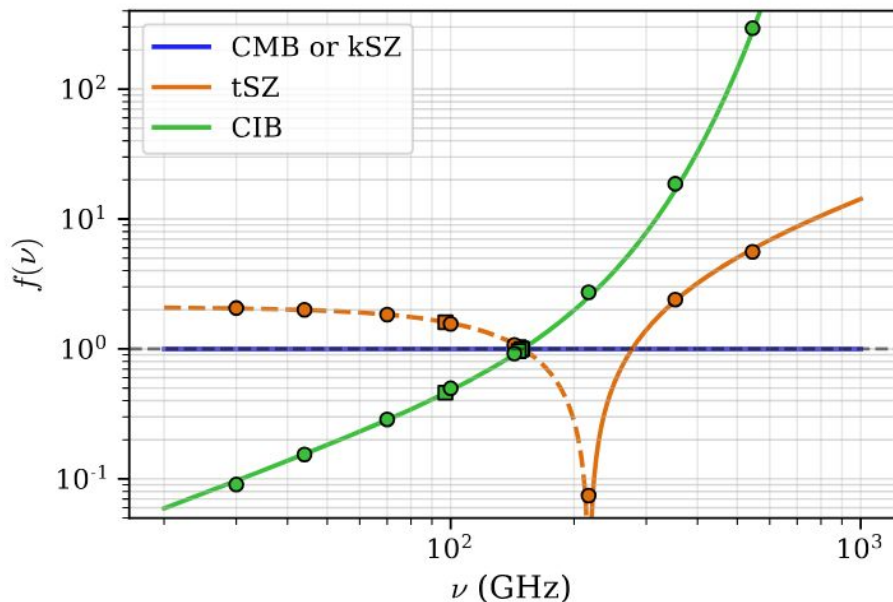
Dominant component
distribution. Calibration
(units = Jy/sr = W·m⁻²)

For small anisotropies

So fluctuations in the

CMB maps are usually
thus independent of
to convert back to Jy.

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}.$$



“temperature” units,
more meaningful

ACT+Planck T
R:f090,G:f150,B:f220

13°×7°

ACT+Planck T

R:f090,G:f150,B:f220

Nearby galaxies

Galaxy clusters

Quasars

High-resolution opens up sensitivity
to late time effects!



13°×7°

Detector noise -> Map noise

NET: temperature fluctuation RMS that gives SNR=1 in 1 second of integration

Typical individual detectors on the ground : $300 \mu\text{K s}^{1/2}$ (space: ~10x lower)

$$\sigma_p = \frac{\text{NET}_{\text{arr}}}{\sqrt{t_p}},$$

Detector noise -> Map noise

NET: temperature fluctuation RMS that gives SNR=1 in 1 second of integration

Typical individual detectors on the ground : $300 \mu\text{K s}^{1/2}$ (space: ~10x lower)

$$\sigma_p = \frac{\text{NET}_{\text{arr}}}{\sqrt{t_p}}$$

Time spent in the pixel

$$t_p = t_{\text{eff}} \frac{\Omega_{\text{pix}}}{A}$$

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$$\sigma_p = \frac{\text{NET}_{\text{arr}}}{\sqrt{t_p}},$$

Time spent in the pixel

$$t_p = t_{\text{eff}} \frac{\Omega_{\text{pix}}}{A}$$

Rewrite in terms of survey properties

$$\sigma_p = \frac{\text{NET}_{\text{arr}}}{\sqrt{t_{\text{eff}} \Omega_{\text{pix}} / A}} = \text{NET}_{\text{arr}} \sqrt{\frac{A}{t_{\text{eff}}}} \frac{1}{\sqrt{\Omega_{\text{pix}}}}.$$

Detector noise -> Map noise

NET: temperature fluctuation RMS that gives SNR=1 in 1 second of integration

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Map depth (“sensitivity”) indep. of pixel size

$$\Delta_T \equiv \sigma_p \sqrt{\Omega_{\text{pix}}}$$

$$\Delta_T = \left(\text{NET}_{\text{arr}} \sqrt{\frac{A}{t_{\text{eff}}}} \frac{1}{\sqrt{\Omega_{\text{pix}}}} \right) \sqrt{\Omega_{\text{pix}}} = \text{NET}_{\text{arr}} \sqrt{\frac{A}{t_{\text{eff}}}}$$

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Map depth (“sensitivity”) indep. of pixel size

$$\Delta_T \equiv \sigma_p \sqrt{\Omega_{\text{pix}}}$$

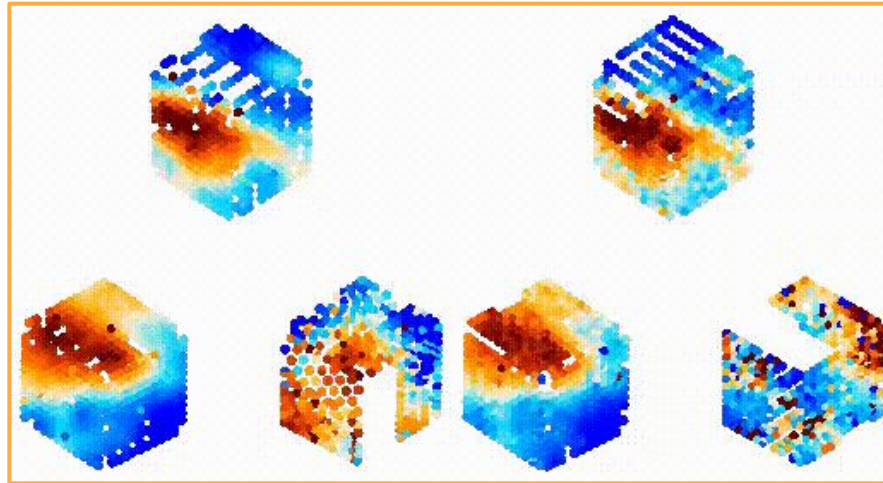
$$\Delta_T = \left(\text{NET}_{\text{arr}} \sqrt{\frac{A}{t_{\text{eff}}}} \frac{1}{\sqrt{\Omega_{\text{pix}}}} \right) \sqrt{\Omega_{\text{pix}}} = \text{NET}_{\text{arr}} \sqrt{\frac{A}{t_{\text{eff}}}}$$

Has the same units ($\mu\text{K-arcmin}$) as sqrt(angular power). Interpret as a standard deviation.

All of this applies to white noise! Correlated atmospheric noise does not average down with time and detector counts in the same way.

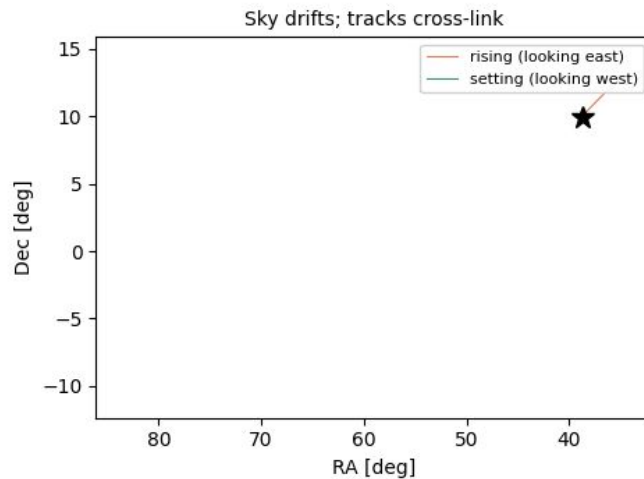
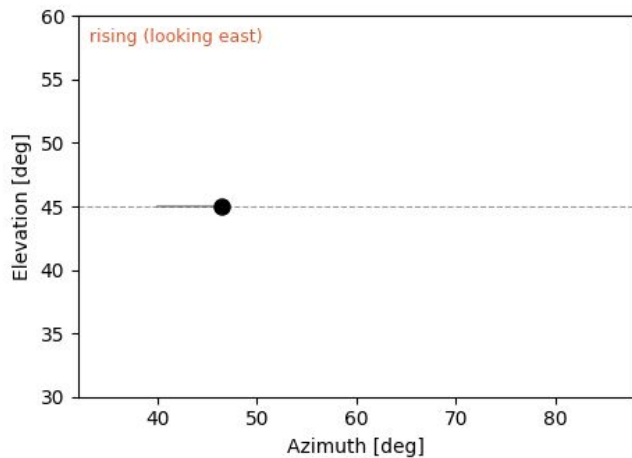
The atmosphere is a source of correlated noise

Clouds moving across the focal plane



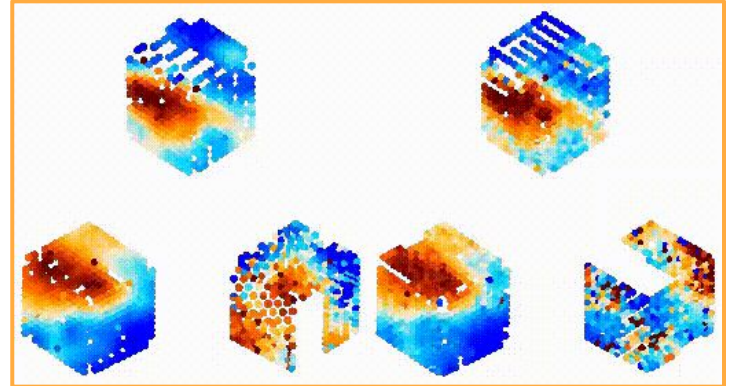
Telescope scans

Constant-elevation scans: horizon vs. sky coordinates



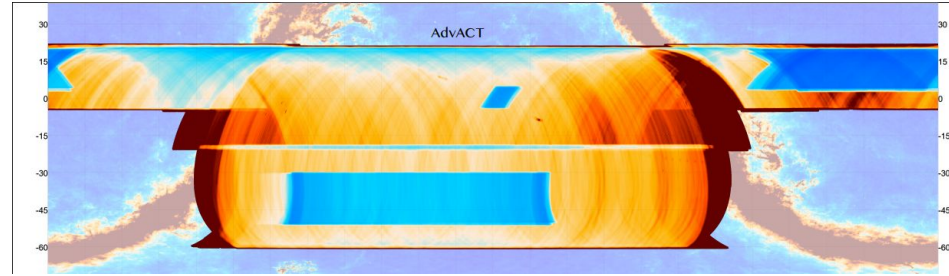
Noise properties

- Correlated (mostly due to the atmosphere)



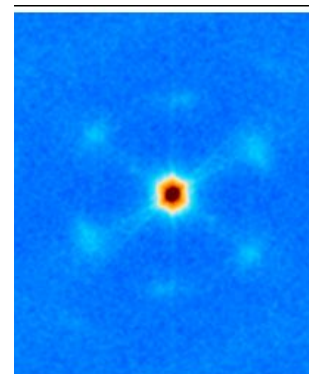
Noise properties

- Correlated (mostly due to the atmosphere)
- Inhomogeneous (due to hits variation)



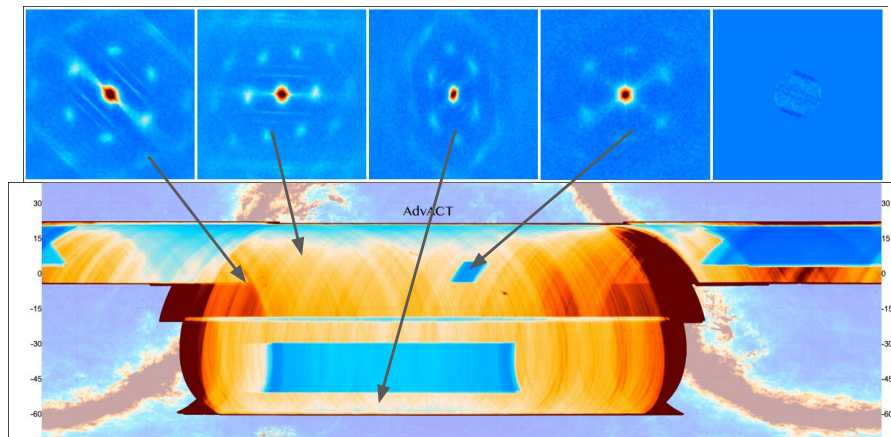
Noise properties

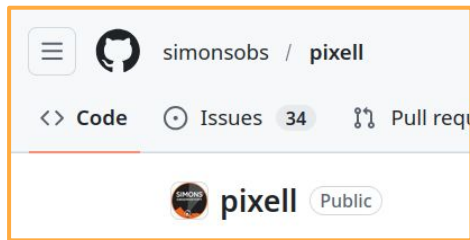
- Correlated (mostly due to the atmosphere)
- Inhomogeneous (due to hits variation)
- Anisotropic (2d Fourier space)
 - Each scan direction has a perp. direction plane wave -> DC mode
 - Cross-linking alleviates this



Noise properties

- Correlated (mostly due to the atmosphere)
- Inhomogeneous (due to hits variation)
- Anisotropic (2d Fourier space)
 - Each scan direction has a perp. direction plane wave -> DC mode
 - Cross-linking alleviates this
- Inhomogeneity of anisotropy
 - Cross-linking varies across the map





pixell and Rectangular* Pixelizations

<https://github.com/simonsobs/pixell/>

Mostly written by **Sigurd Naess** ([@amaurea](#)) + contributions from other ACTors
We use it extensively in ACT and SO!

*Image plane is a rectangular grid, 2d array

What pixell can do

Manipulate maps in rectangular pixelization

2D Fourier transforms

Spherical harmonic transforms

Reproject and interpolate between different pixelizations, including healpix

Simulate Gaussian maps

Lens, modulate and aberrate CMB maps

Simulate, find and subtract point sources and clusters

+++

What pixell can't/won't do

Pseudo-spectra with mode decoupling

(see [namaster](#), [pspy](#), [wiggles](#), etc.)

Lensing reconstruction

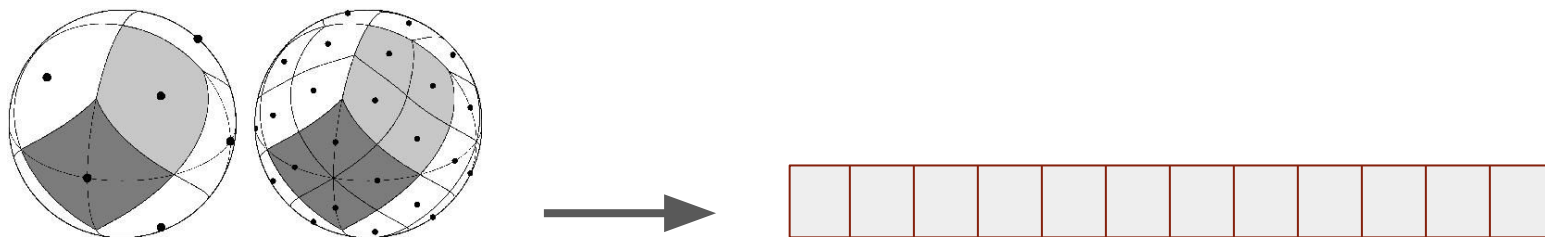
(see [sym lens](#), [falafel](#), etc.)

+++

Basics: contrasting with Healpix

We observe the sky (a sphere), but to map it and analyze it -> represent in pixels in computer memory

Healpix: Used by Planck, WMAP, etc. scheme for mapping equal area divisions of the sphere on to a **1-dimensional array**



1-dimensional array*

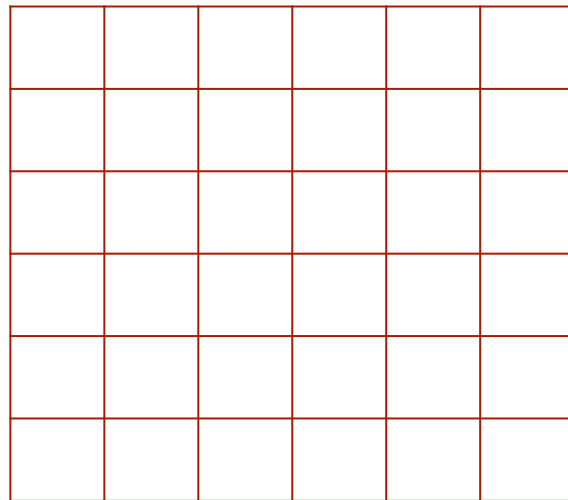
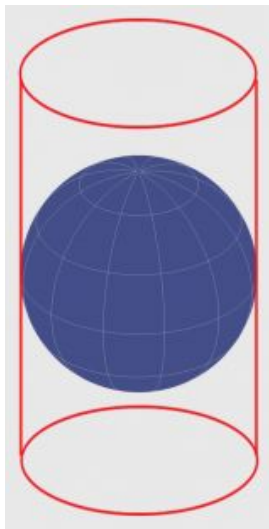
* Can be represented in 2d, but looks weird and has empty regions, depends on NESTED vs. RING, not contiguous in memory. Makes interpolation hard.

Basics: rectangular pixelization

We observe the sky (a sphere), but to map it and analyze it -> represent in pixels in computer memory

Rect. pix: Used widely in geophysics and astronomy, including in ACT and SPT CMB maps; use one of many possible map projection schemes but map to **2D array**.

e.g. cylindrical (but many other projections possible)



2-dimensional array

Healpix maps can be analyzed
with: ***healpy***

Rect.pix. maps can be
analyzed with: ***pixell***

<https://github.com/simonsobs/pixell/>

Healpix maps are defined by
nside
(powers of 2 in basic implementation)

pixell rect. pix. maps are
defined by a **shape, wcs** pair

Rect. pix. is unified by the FITS
World Coordinate System (WCS)
standard.

Basics: World Coordinate System

```
SIMPLE = T / conforms to FITS standard
BITPIX = -64 / array data type
NAXIS = 2 / number of array dimensions
NAXIS1 = 3521
NAXIS2 = 1505
CTYPE1 = 'RA---CEA'
CTYPE2 = 'DEC--CEA'
CUNIT1 = ' '
CUNIT2 = ' '
CRVAL1 = 0
CRVAL2 = 0
CDELT1 = -0.008333333333333333
CDELT2 = 0.008333333333333333
CRPIX1 = 5832
CRPIX2 = 1302
PV2_1 = 1
EQUINOX = 2000
PC1_1 = 1
PC1_2 = 0
PC2_1 = 0
PC2_2 = 1
-----
```

WCS standard defines set of key-values in a FITS file header (or in a Python dictionary) that gives us a unique prescription for mapping

pixel index in 2D grid

<->

sky coordinate

This gives us all the info we need to analyze a map and make physical inferences from it.

Misconception: Rect.pix. is “flat-sky”

- Both healpix and rect.pix. have an information preserving (up to pixel averaging) scheme for mapping sky coordinates (on the **sphere**) to memory locations
- The fact that rect.pix. lays it out on a 2D grid does **not** in any way make it a “flat sky representation”. Healpix lays it out in 1D, so if that were true, it’s even... flatter
- Whether or not you make a “flat sky approximation” depends on the details of what you do with those pixels: a common flat-sky approximation is using 2d FFTs instead of spherical harmonics.
- Rect.pix. maps can be analyzed with spherical harmonics: *pixell* has interfaces with *ducc0* for that

Rect.pix.: specializing to cylindrical projections

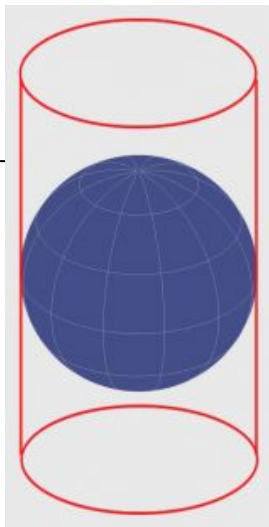
OLD ACT RELEASES

5.2.2. CEA: Cylindrical equal area

The cylindrical equal area projection is the special case of the cylindrical perspective projection with $\mu = \infty$. It is conformal at latitudes $\pm\theta_c$ where $\lambda = \cos^2 \theta_c$. The formulæ are

$$x = \phi,$$
$$y = \frac{180^\circ}{\pi} \frac{\sin \theta}{\lambda},$$

(79)



These systems are a bit weird at the poles, but in practice this does not cause any problems even for full-sky maps.

CURRENT ACT / FUTURE RELEASES

5.2.3. CAR: Plate carrée

The equator and all meridians are correctly scaled in the plate carrée projection¹⁴, whose main virtue is that of simplicity. Its formulæ are

$$x = \phi, \tag{83}$$

$$y = \theta. \tag{84}$$

The projection is illustrated in Fig. 18.

Calabretta, Greisen, 2002
<https://arxiv.org/abs/astro-ph/0207413>

Visualizing scans in a cylindrical projection

pixell abstracts away projections
so you don't have to think about
e.g.

CEA (old ACT)

CAR (current and future ACT/SO)

ZEA (SPT)

e.g. many routines in the [CMB Summer School notebooks](#) are generalized and encapsulated into easy-to-use functions

Healpix and rect.pix. have their pros and cons

*For **ground-based high-resolution** experiments, these weigh strongly in favor of rect. pix.*

Healpix vs. CAR

	Healpix w/ healpy	CAR w/ pixell
Spherical harmonics	😐	✓
Cut-sky (partial sky maps)	✗	✓
Interpolation (e.g. lensing, postage stamps)	😐	✓
Visualization and exploration	😐	✓
sky2pix (e.g. map-making)	😐	✓
2D Fourier basis (ground-based noise)	✗	✓
Equal area	✓	✗
Overpixelization	😐	😐

Secondaries

CMB Gravitational Lensing

Image: ESA/Planck

Primordial CMB (most distant and oldest source of radiation)

Distance/redshift to CMB source well known, matter mildly nonlinear

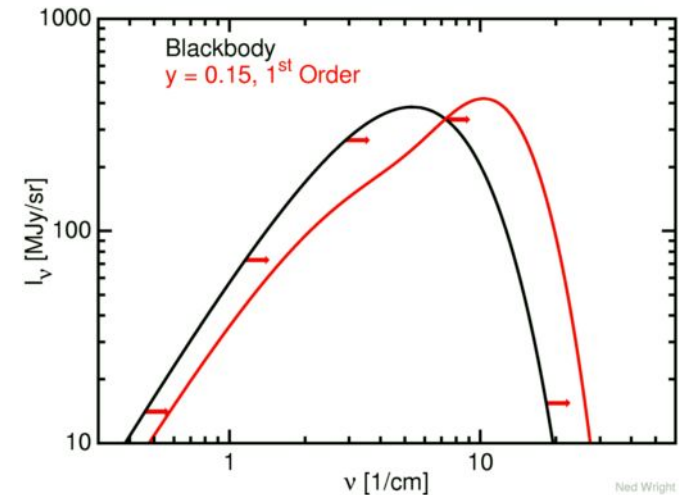
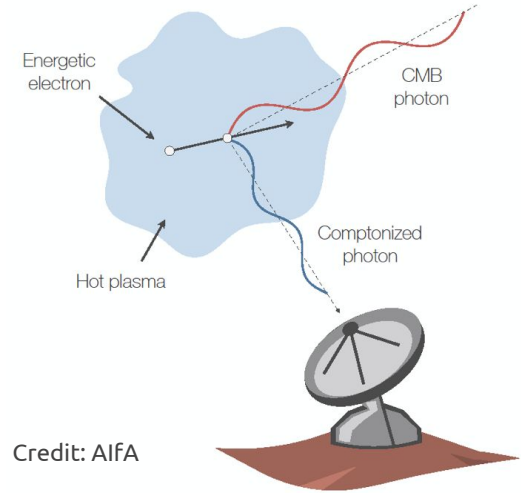
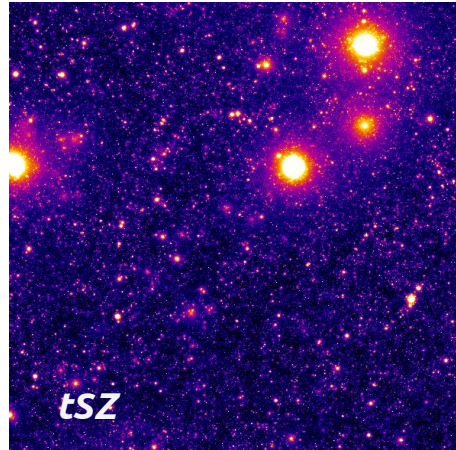
Distribution of mass deflects CMB light that passes through

Thermal Sunyaev-Zel'dovich Effect

Inverse Compton
scattering of CMB
photons off hot clouds
of free electrons

Probes free electron
thermal pressure
(hot gas)

y-distortion to photon
spectrum.



Thermal Sunyaev-Zel'dovich Effect

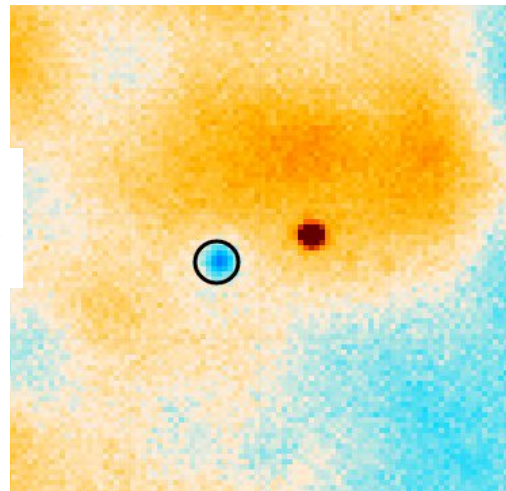
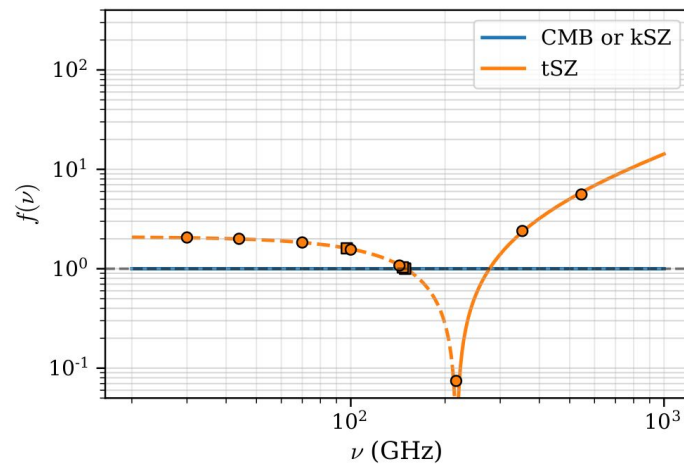
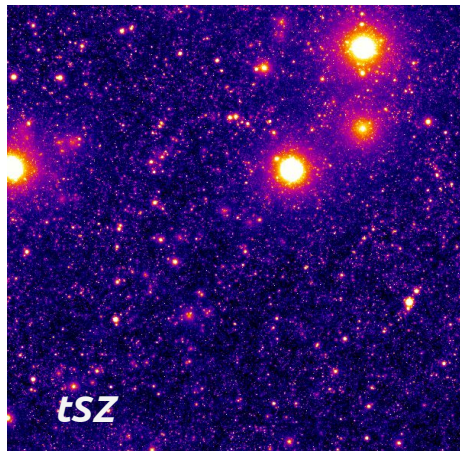
Inverse Compton
scattering of CMB
photons off hot clouds
of free electrons

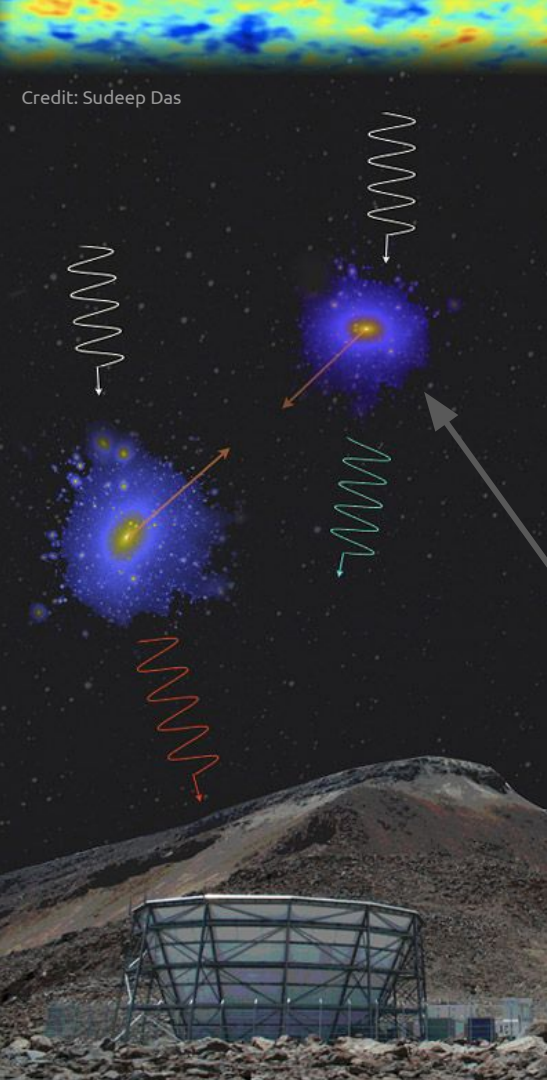
Probes free electron
thermal pressure
(hot gas)

y-distortion to photon
spectrum.

$$\frac{\delta T_{\text{tSZ}}}{T_{\text{CMB}}} = f(\nu) \tau \left(\frac{v_{\text{thermal}}}{c} \right)^2 \propto \tau T_e$$

$$\tau \sim 10^{-3}, \quad \frac{v_{\text{thermal}}}{c} \sim 0.1$$





kSZ (kinetic Sunyaev Zeldovich) effect: Doppler shift of CMB photons scattering off electrons with bulk velocity

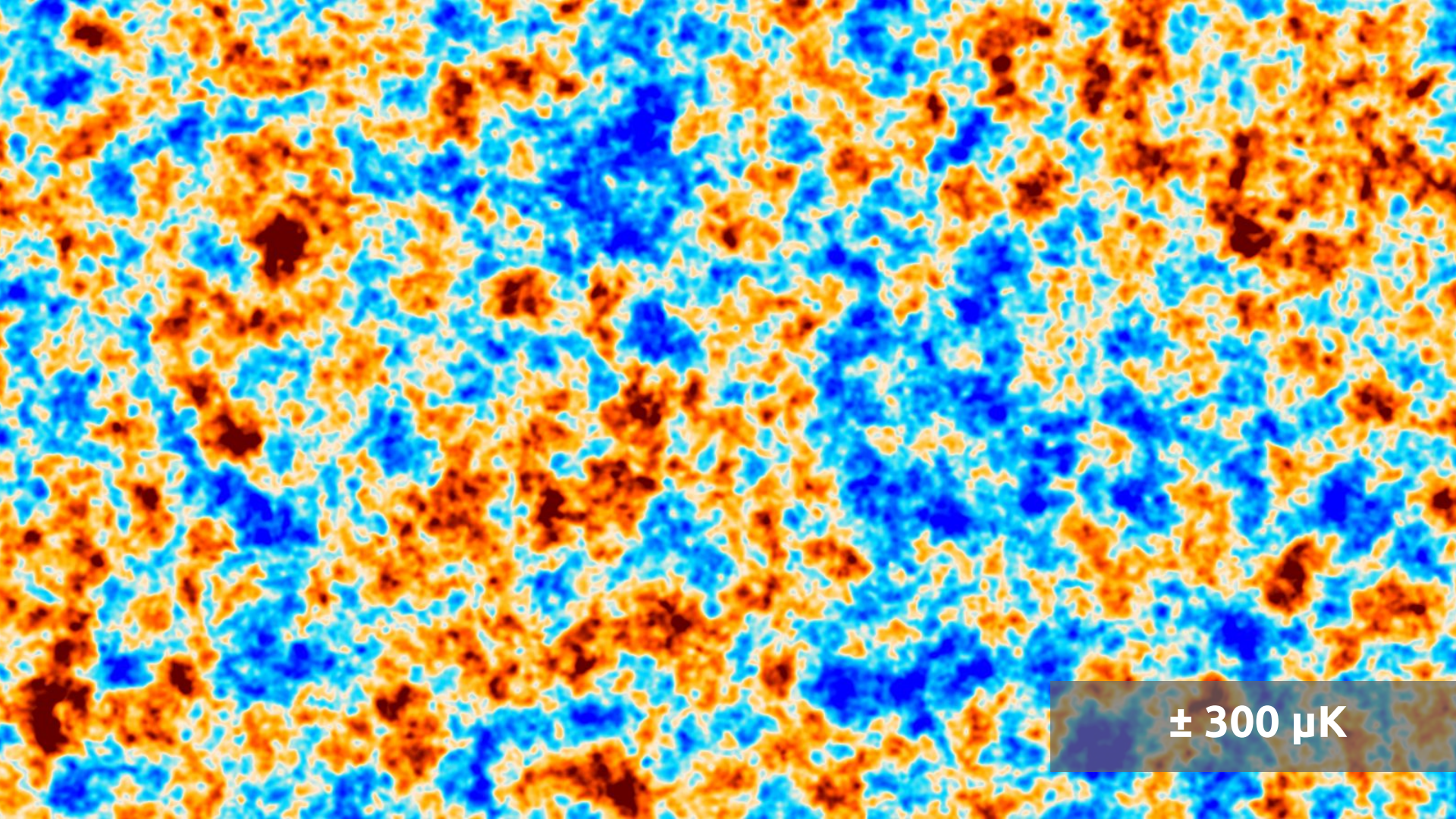
$$\frac{\Delta T_{\text{kSZ}}(\vec{\mathbf{n}})}{T_{\text{CMB}}} \sim \int d\chi e^{-\tau(z)} v_r \delta_e(\vec{\mathbf{n}}, \chi)$$

**Ionized gas in and
between galaxy clusters**

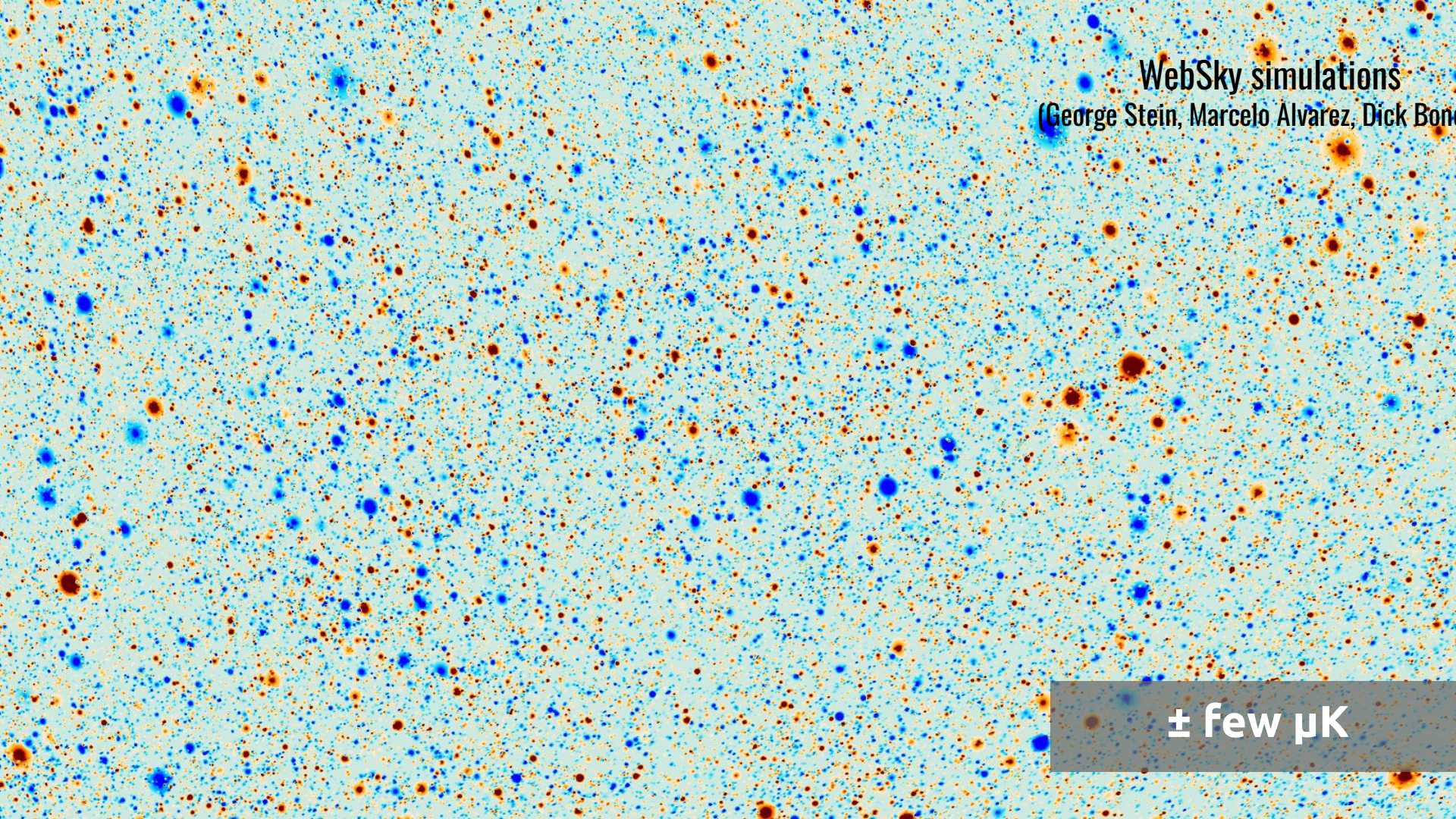
Applications include robust
primordial
non-Gaussianity

Currently detected only at the
<20 σ level

But expected to improve
quickly with (deeper) CMB
and larger galaxy volume.
SNR O(100-1000) expected!



$\pm 300 \mu\text{K}$

The background of the slide is a dense field of small, multi-colored dots in shades of blue, orange, and yellow, representing a simulated sky map. The dots are distributed across the entire frame, with some appearing as larger, more prominent spots.

WebSky simulations
(George Stein, Marcelo Alvarez, Dick Bond)

$\pm$ few μK

Moving lens

$$\frac{\delta T(\hat{n})}{T_{\text{CMB}}} = -\frac{2}{c^3} \int_{\chi_0}^{\chi_*} \nabla_{\perp} \Phi(\hat{n}, \chi) \cdot \mathbf{v}_t(\hat{n}, \chi) d\chi$$
$$\sim 0.2 \mu\text{K} \left(\frac{M_{200}}{10^{14} M_{\odot}} \right)^{2/3} \left(\frac{v_{\perp}}{500 \text{ km s}^{-1}} \right)$$

CMB photon lensed by different part of potential as it enters and exits moving lens halo.

Dipole-like signal probing transverse velocity field

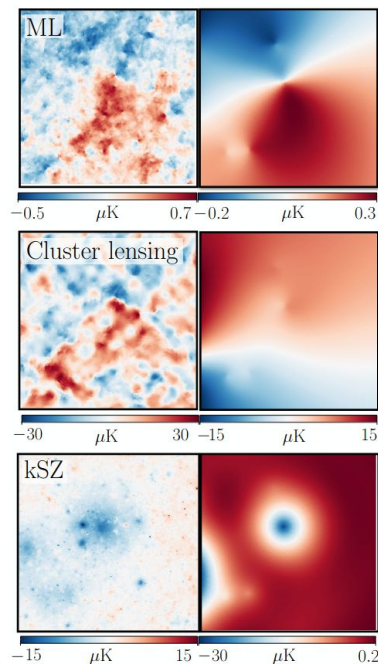
Hotinli et al 2026 recent detection

Estimator

Hotinli et al 2019

Cosmological applications

Hotinli, Smith, MM et al 2021



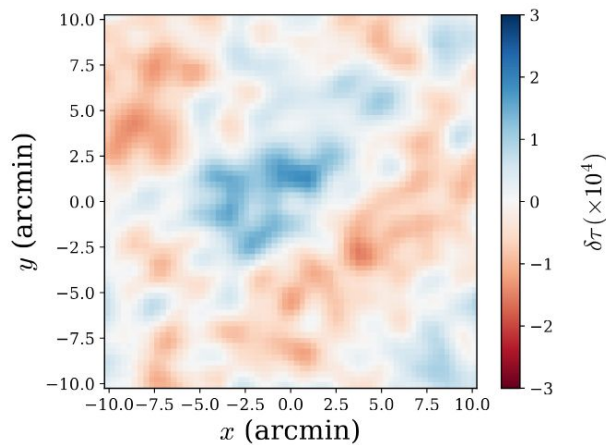
Patchy screening

$$\Delta T_{\text{prim}}(\hat{\mathbf{n}}) \rightarrow e^{-\tau(\hat{\mathbf{n}})} \Delta T_{\text{prim}}(\hat{\mathbf{n}}) \approx [1 - \delta\tau(\hat{\mathbf{n}})] \Delta T(\hat{\mathbf{n}})$$

CMB photons get scattered into or out of LOS by Thompson scattering off ionized electrons. Proportional to optical depth, but suppressed by primary anisotropy $\sim 10^{-5}$.

Another probe of ionized gas!

Detectability strongly depends on other foregrounds, so simulations important!



1.7 sigma constraint from *Coulton et al. 2024*

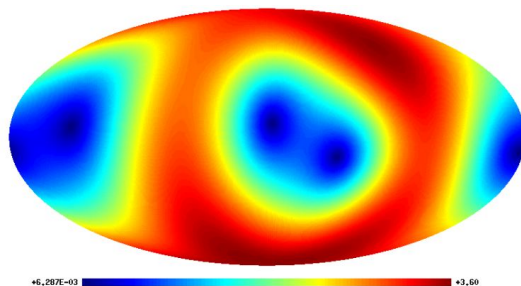
See also *Schutt et al 2024*

Polarized SZ

- CMB photons scatter off ionized gas; quadrupole leads to polarization just like at recombination
- Should be detectable with future CMB; helps go beyond cosmic variance for CMB quadrupole

$$(Q + iU) \propto \tau a_{2m}(\mathbf{r}_{\text{cluster}})$$

$$P_{\text{quad}} \sim 5 \text{ nK} \left(\frac{\tau}{10^{-3}} \right)$$



Correlated polarization magnitude at $z=0.05$

Hall, Challinor 2014

See also Cayuso, Johnson 2020

Rotational kSZ: the Baxter effect

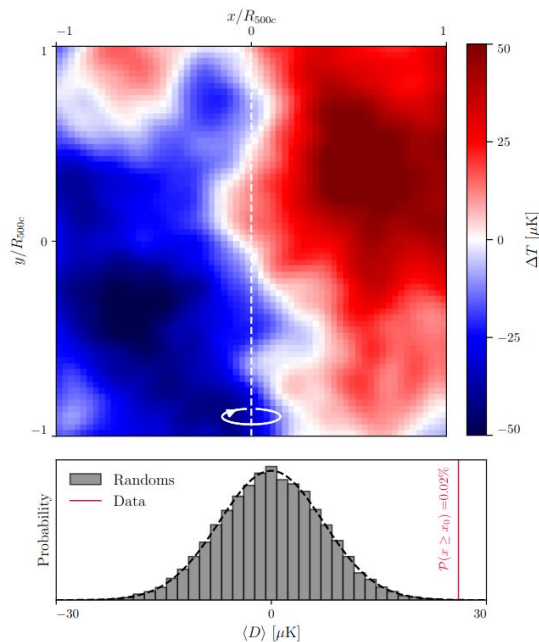
Rotations (but also mergers, substructure) etc. contribute to dipole like kSZ signal. 10-100 μK in large clusters. Source of noise that is not in many sims!

Hint seen in:

Baxter, Sherwin, Raghunathan 2019

Detected at 3.8 sigma (with E-nulling!)

Goldstein, Hill 2025



Stacking analyses

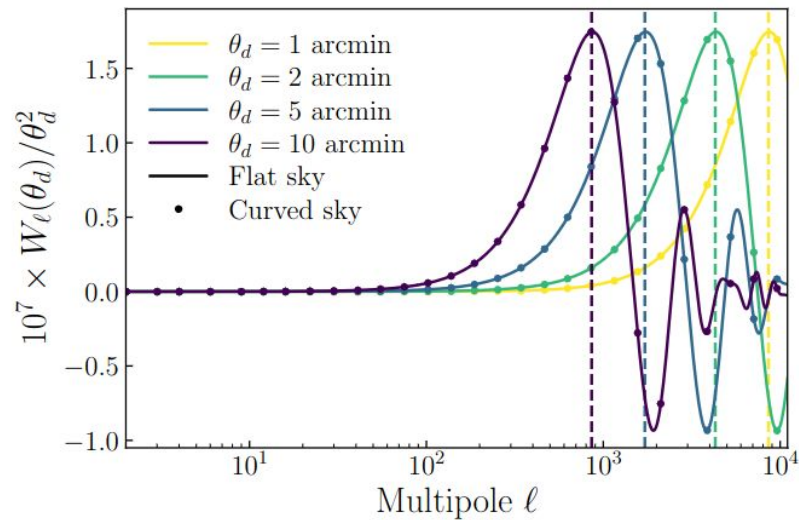
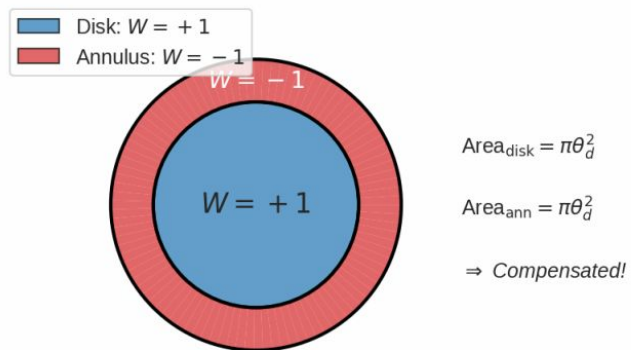
Stacking sources

- Stacking typically means projecting a small region (thumbnail) to a **tangent plane**, then averaging
- For radially symmetric objects, compress info into some statistic (radial profile, CAP): **the bins will be correlated!**
 - large offset in raw radial profile due to correlated noise (CMB and atmosphere)
- Most works estimate covariance from the sample itself (e.g. cov of mean, jackknife, bootstrap): must be checked with sims for dense samples since independence breaks down!

CAP filters

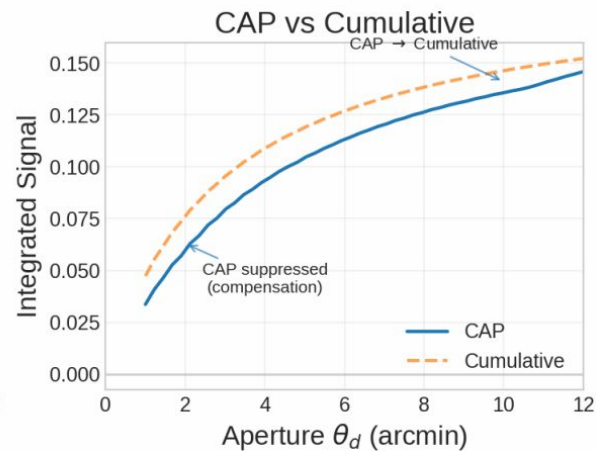
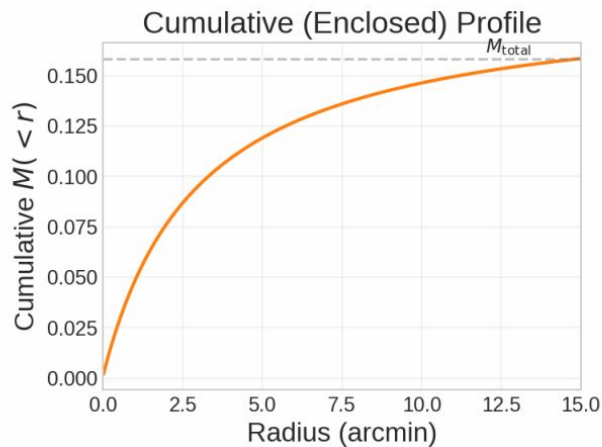
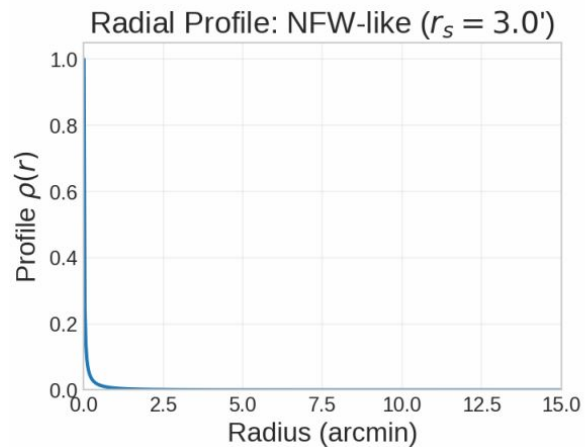
$$\tilde{W}_{\theta_d}(\ell) = \frac{4\pi\theta_d^2}{x} \left[J_1(x) - \frac{1}{\sqrt{2}} J_1(\sqrt{2}x) \right]$$

CAP Filter: Real Space

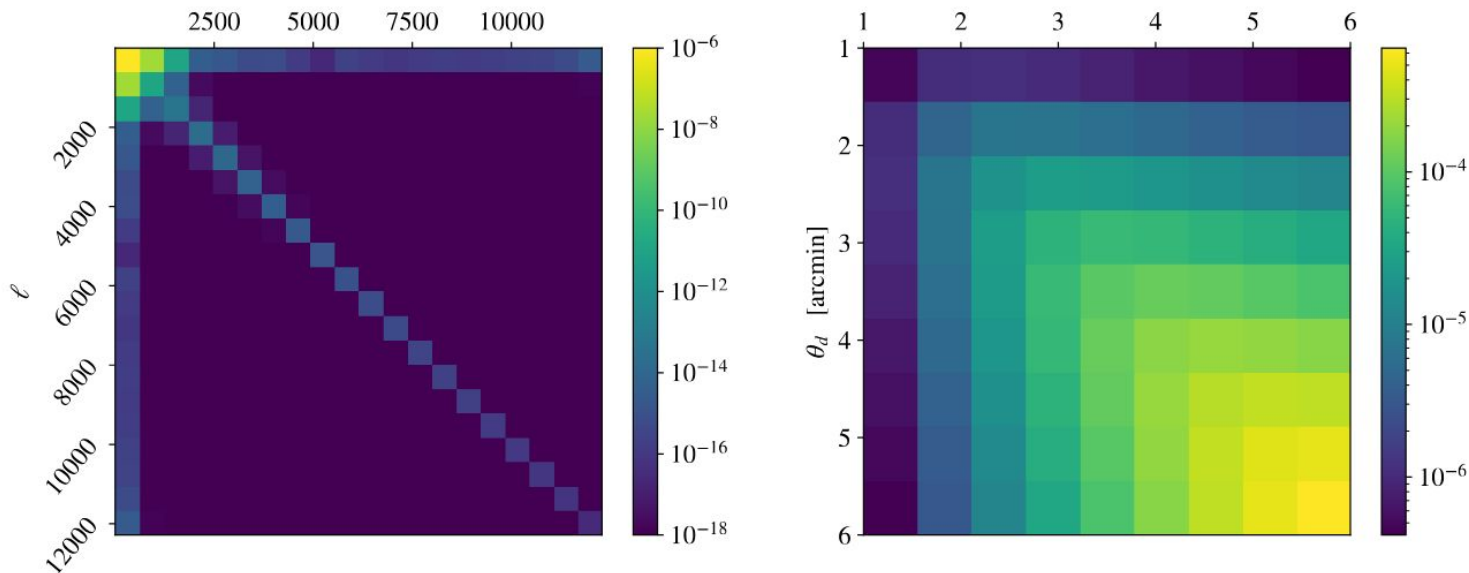


Harscouet et al (arXiv:2512.14625)

CAP filters

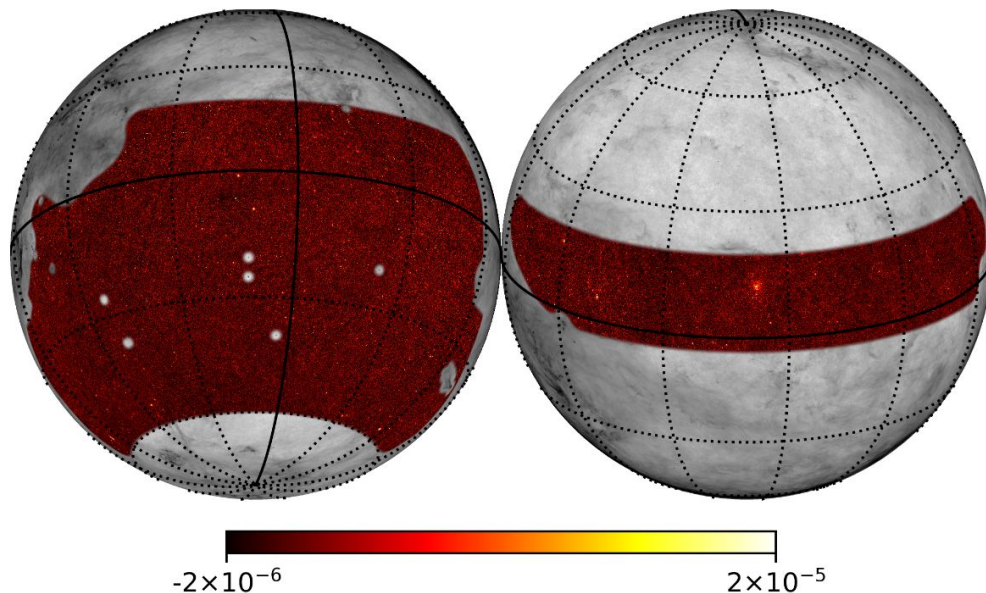


CAP filters



Harscouet et al (arXiv:2512.14625): Fourier kSZ as an alternative to CAP

Compton- γ (and other ILC) maps



ILC maps just sum up individual frequency maps with weights (minimizing noise, unity response to component of interest, optional null response to contaminant)

$$M(\ell) = \sum_i w_i(\ell) M_i(\ell)$$

$$M_i(\ell) = a_i(\ell)s(\ell) + a'_i(\ell)c(\ell) + n_i(\ell)$$

$$w_i = a_j C_{ji}^{-1} / (a_j C_{jk}^{-1} a_k)$$

$$w_i = \frac{\left(a'_j C_{jk}^{-1} a'_k \right) a_j C_{ji}^{-1} - \left(a_j C_{jk}^{-1} a'_k \right) a'_j C_{ji}^{-1}}{\left(a_j C_{jk}^{-1} a_k \right) \left(a'_j C_{jk}^{-1} a'_k \right) - \left(a_j C_{jk}^{-1} a'_k \right)^2}$$

Standard: minimizes total power -> foreground residuals can still be large!

Constrained: deprojects SED

SED could be wrong! Many populations with many SEDs!

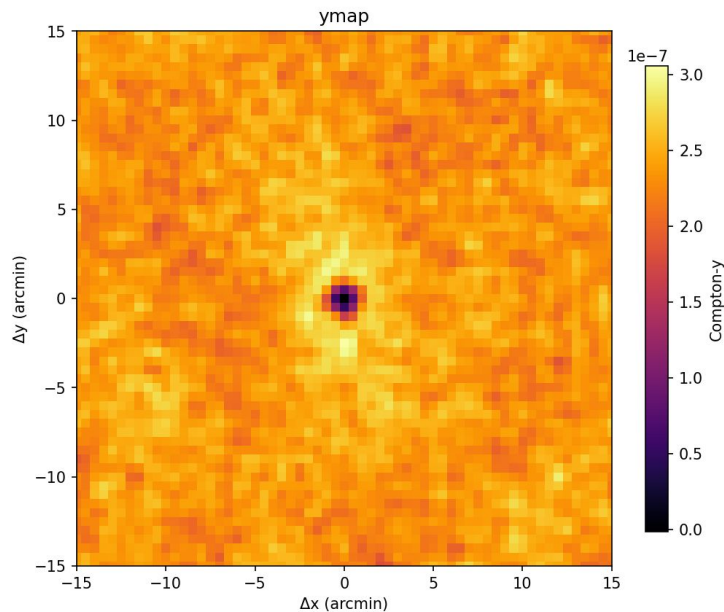
(Use moment deprojection, but get hit with noise: [Chluba, Hill, Abitbol arxiv:1701.00274](#))

Deprojecting one component can amplify another

Moral: test on simulations. Maybe consider multi-frequency fitting instead.

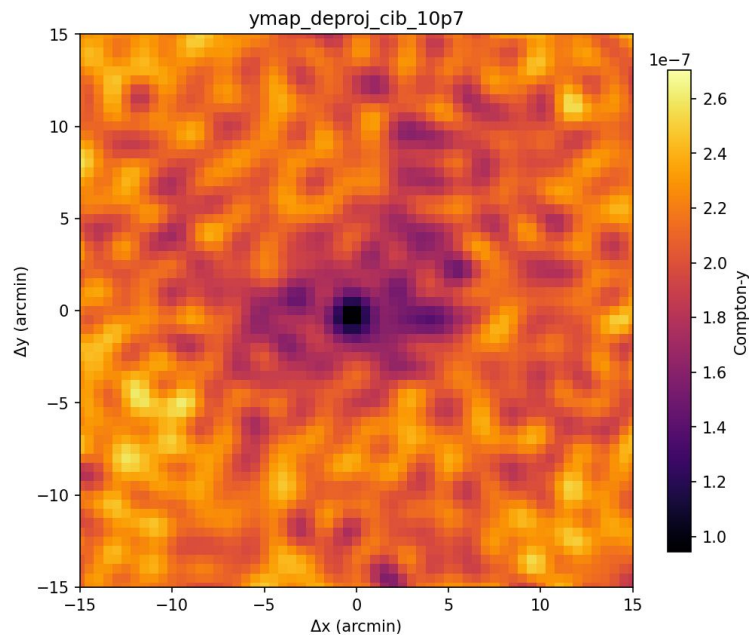
Extreme example of things going wrong

“Negative” Compton- y from eBOSS quasars



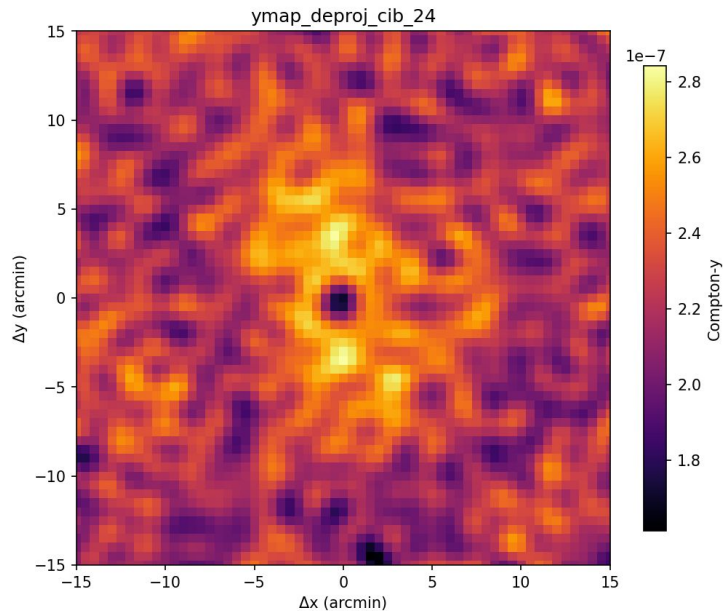
Extreme example of things going wrong

Deproject dust SED?



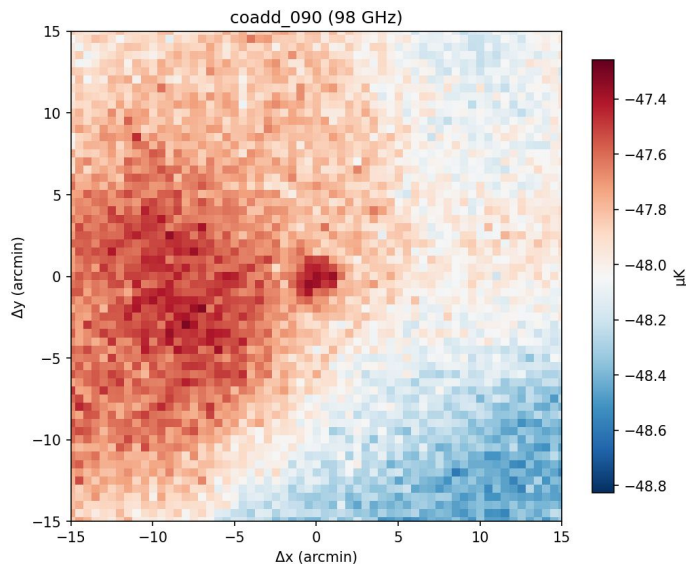
Extreme example of things going wrong

Deproject a different dust SED?



Extreme example of things going wrong

Emission at 90 GHz means radio component



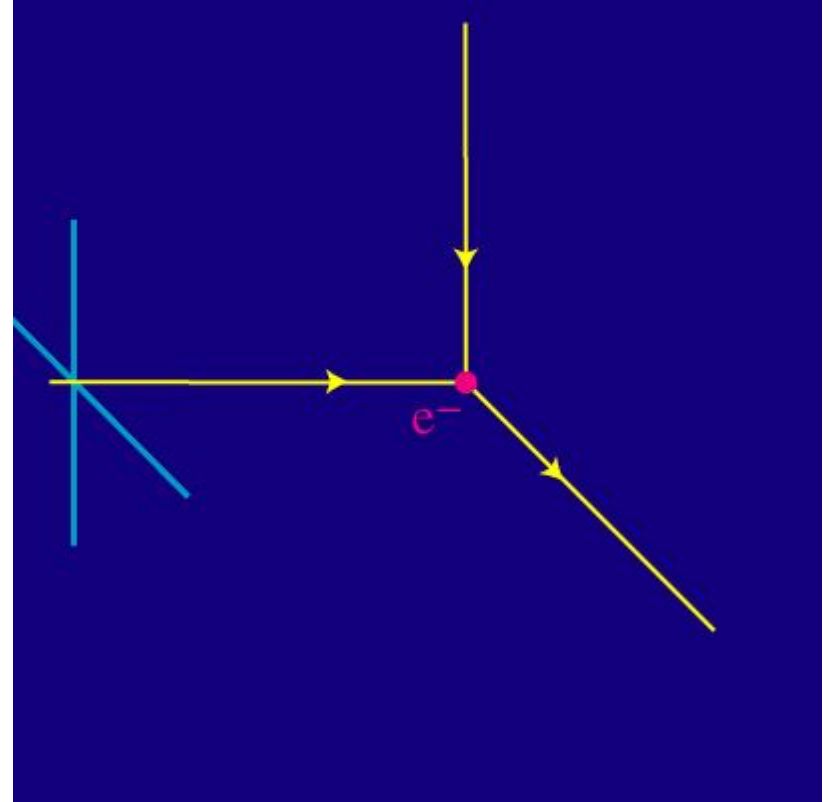
Multi-frequency fit is probably the way to go, not Compton-y

Polarization

Scattering leads to linear polarization

Wayne Hu

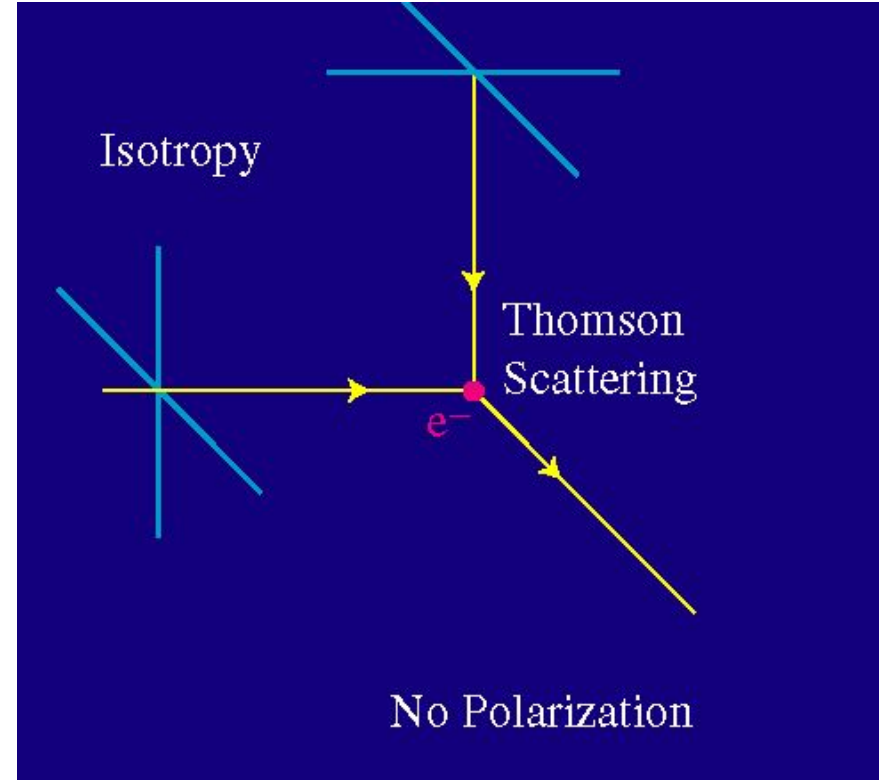
- CMB photons (~ 0.26 eV at recomb.) Thomson scatter off electrons (0.5 MeV)
- Photons (massless) cannot be polarized along the direction of their propagation
- Shown here: unpolarized photon scattering at 90 deg



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Wayne Hu

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- With isotropic radiation, it's a wash

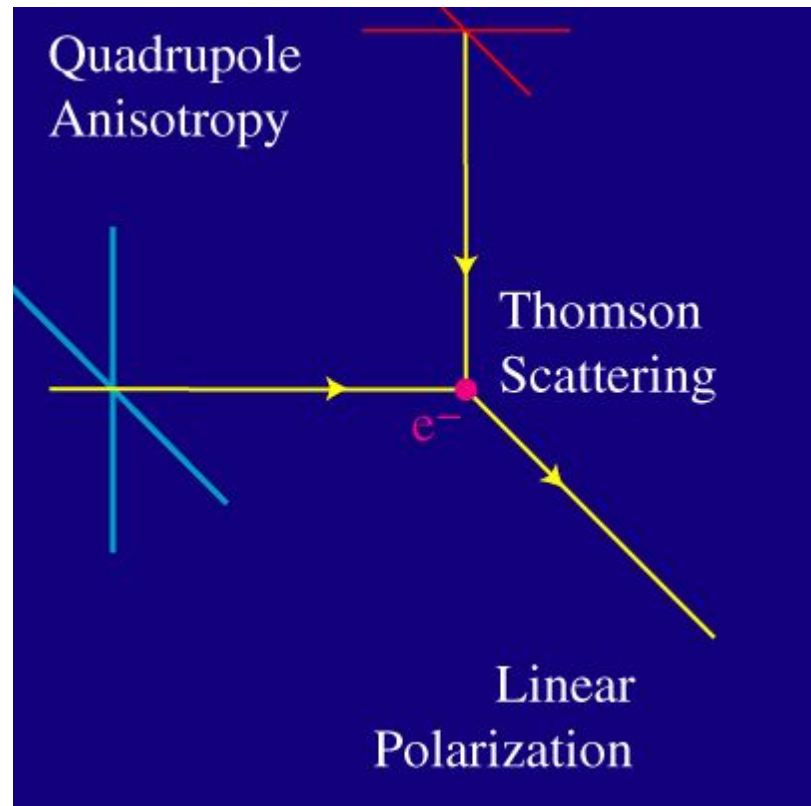


Scattering leads to linear polarization

Wayne Hu

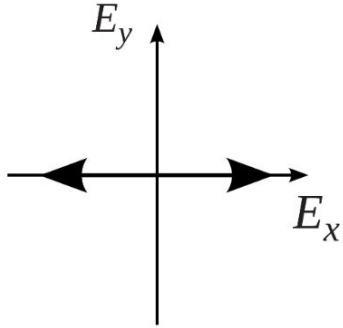
- CMB photons (~ 0.26 eV at recomb.) Thomson scatter off electrons (0.5 MeV)
- Photons (massless) cannot be polarized along the direction of their propagation
- Shown here: unpolarized photon scattering at 90 deg
- With isotropic radiation, it's a wash
- The only part of the radiation that sources net polarization is the quadrupole moment

$$\frac{d\sigma_T}{d\Omega} = \frac{3\sigma_T}{8\pi} |\hat{\epsilon}' \cdot \hat{\epsilon}|^2.$$



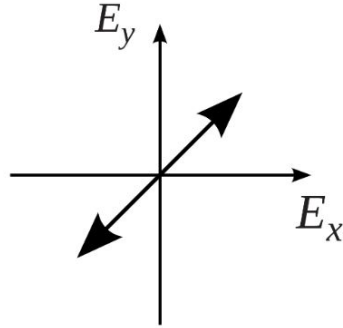
Stokes Vectors

$$S = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} |E_x|^2 + |E_y|^2 \\ |E_x|^2 - |E_y|^2 \\ 2\mathbf{Re}(E_x E_y e^{i(\delta_x - \delta_y)}) \\ -2\mathbf{Im}(E_x E_y e^{i(\delta_x - \delta_y)}) \end{bmatrix}$$



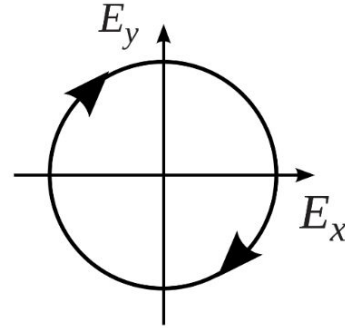
Horizontal

$$I = +Q \\ U = V = 0$$



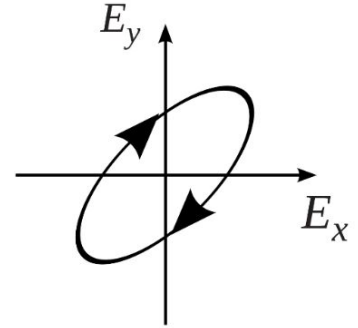
$\pm 45^\circ$

$$I = +U \\ Q = V = 0$$



Circular

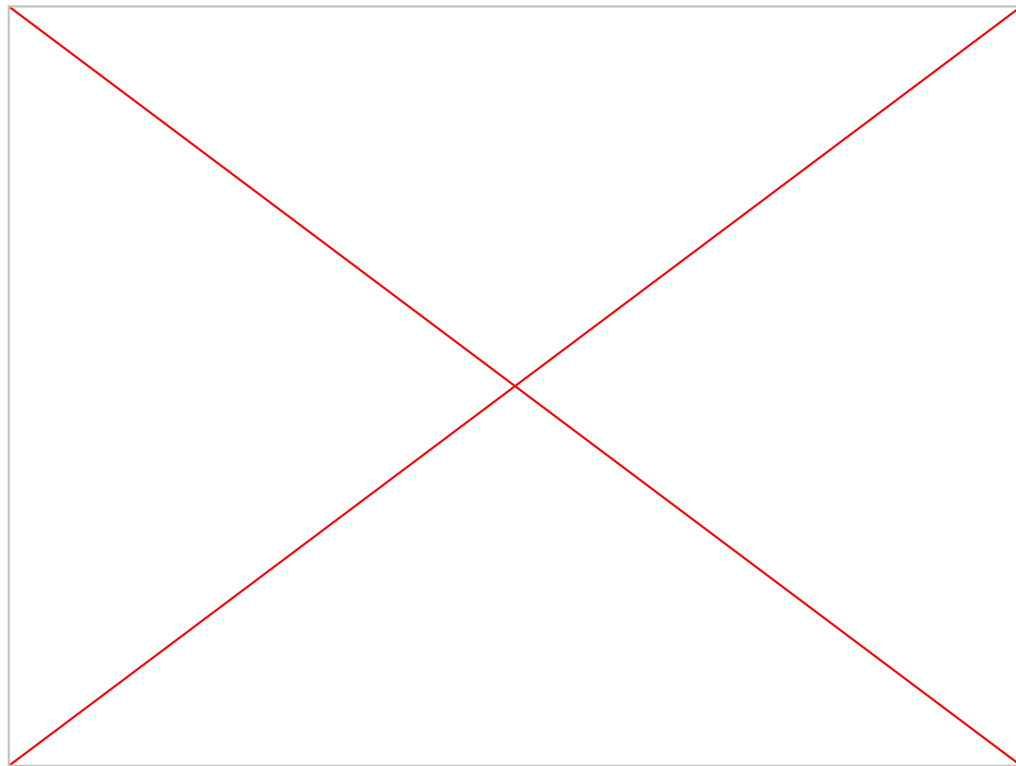
$$I = +V \\ Q = U = 0$$



Elliptical

$$I = \sqrt{Q^2 + U^2 + V^2} \\ Q \neq U \neq V$$

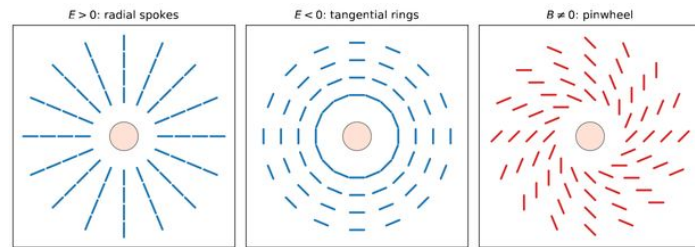
Vector fields vs. spin-2



Coordinate independent representation of polarization

$Q+iU$ transforms as a spin-2

$$P_{ij}(\mathbf{x}) = \begin{pmatrix} Q & U \\ U & -Q \end{pmatrix}.$$



Another Helmholtz / SVT decomposition

$$P_{ij} = (\partial_i \partial_j - \frac{1}{2} \delta_{ij} \nabla^2) E + \frac{1}{2} (\epsilon_{ik} \partial_k \partial_j + \epsilon_{jk} \partial_k \partial_i) B.$$

$$Q(\mathbf{x}) \pm i U(\mathbf{x}) = \frac{1}{2} (\partial_x \pm i \partial_y)^2 [E(\mathbf{x}) \mp i B(\mathbf{x})],$$

Transform to 2D Fourier space

$$[\tilde{Q}(\ell) + i \tilde{U}(\ell)] e^{-2i\phi_\ell} = E_\ell + i B_\ell,$$

How do pure-E and
pure-B modes look for
plane waves?

E/B patterns

$$Q(\mathbf{l}) = E(\mathbf{l}) \cos(2\phi_{\mathbf{l}}) - B(\mathbf{l}) \sin(2\phi_{\mathbf{l}})$$

$$U(\mathbf{l}) = E(\mathbf{l}) \sin(2\phi_{\mathbf{l}}) + B(\mathbf{l}) \cos(2\phi_{\mathbf{l}})$$

Single plane wave

$$Q(\boldsymbol{\theta}) = [E(\mathbf{l}) \cos(2\phi_{\mathbf{l}}) - B(\mathbf{l}) \sin(2\phi_{\mathbf{l}})] e^{i\mathbf{l} \cdot \boldsymbol{\theta}}$$

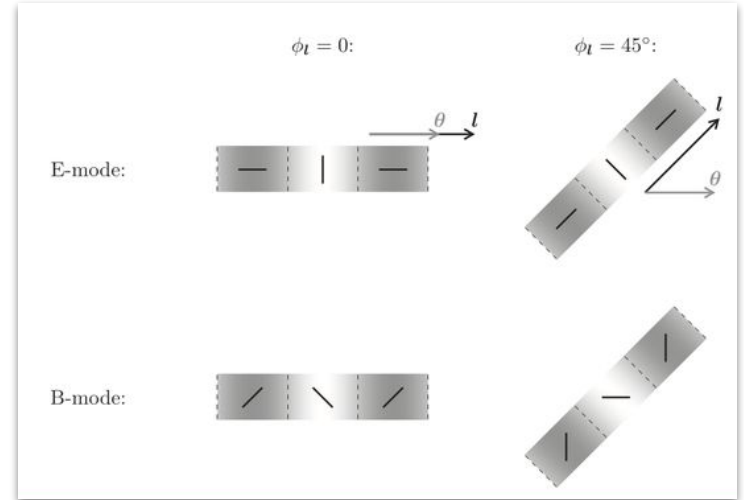
$$U(\boldsymbol{\theta}) = [E(\mathbf{l}) \sin(2\phi_{\mathbf{l}}) + B(\mathbf{l}) \cos(2\phi_{\mathbf{l}})] e^{i\mathbf{l} \cdot \boldsymbol{\theta}}$$

Pure E-mode

$$Q(\boldsymbol{\theta}) = E(\mathbf{l}) \cos(2\phi_{\mathbf{l}}) e^{i\mathbf{l} \cdot \boldsymbol{\theta}}$$

$$U(\boldsymbol{\theta}) = E(\mathbf{l}) \sin(2\phi_{\mathbf{l}}) e^{i\mathbf{l} \cdot \boldsymbol{\theta}}.$$

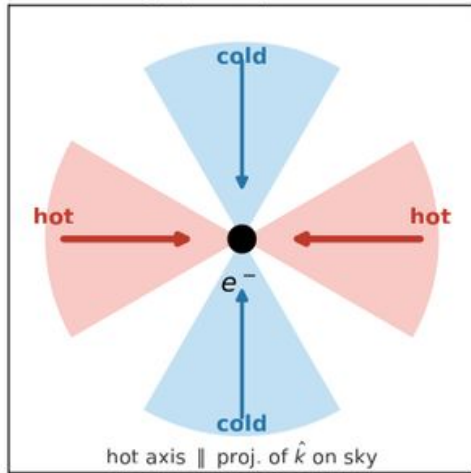
always parallel or perp. to wave-vector



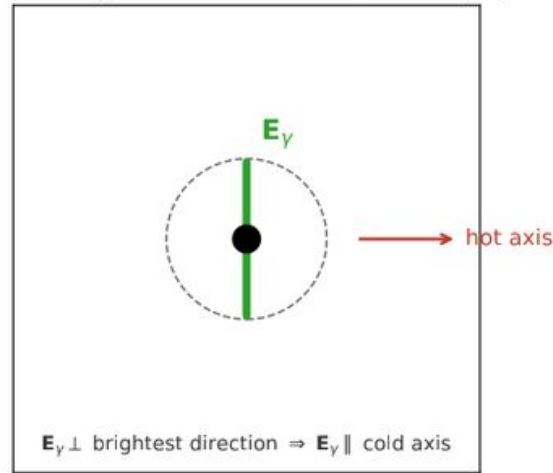
Baumann

Building up to plane wave picture: Thomson scattering produces polarization perpendicular to direction of brightest incoming radiation

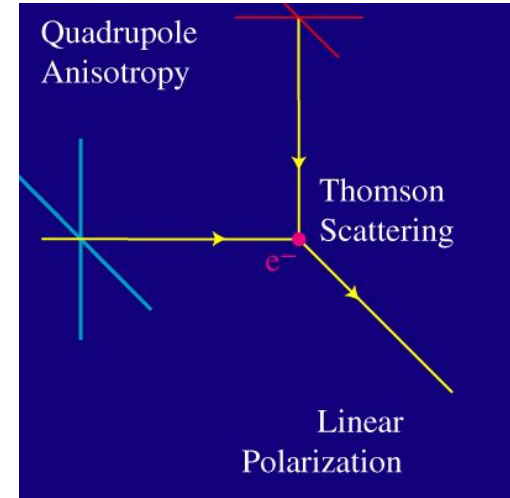
Local intensity quadrupole at the electron



Scattered light toward observer ($\odot$ out of page)

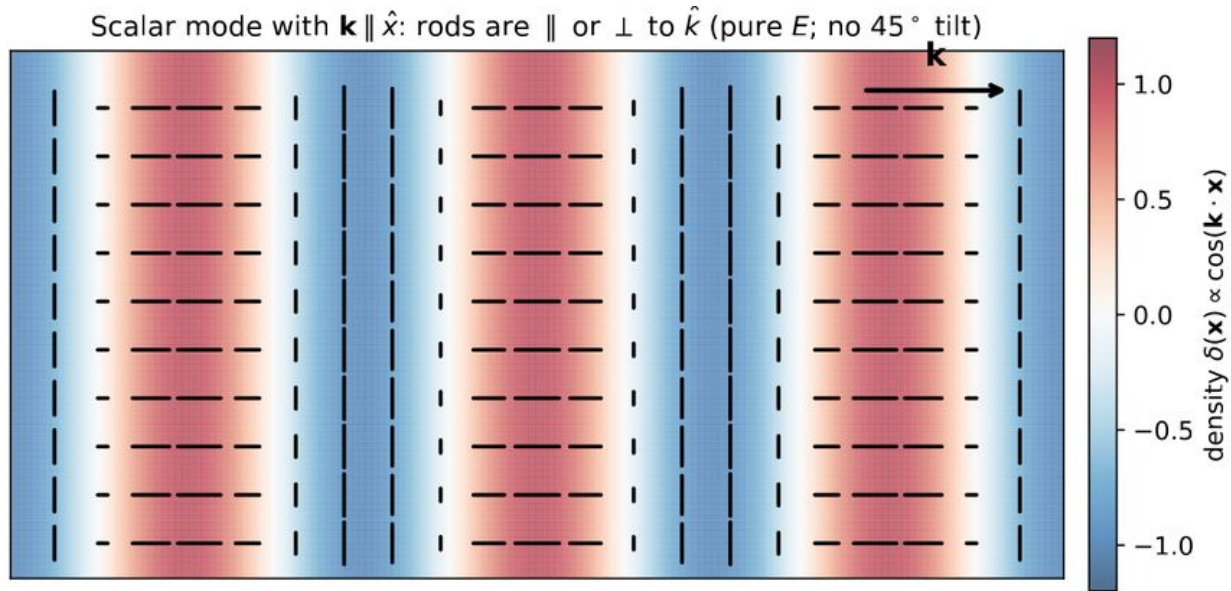


Wayne Hu

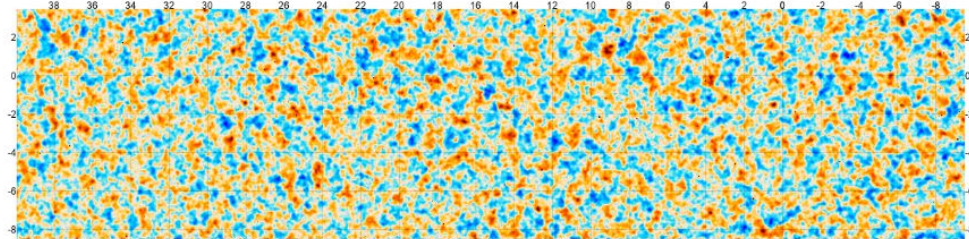


Scalar modes: build from plane waves

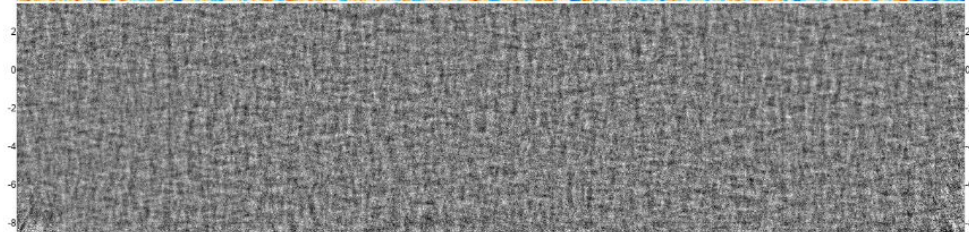
Scalar fluctuations are sums of single plane waves



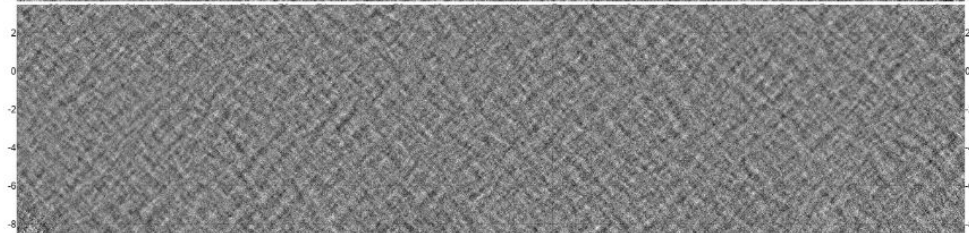
T (or I)



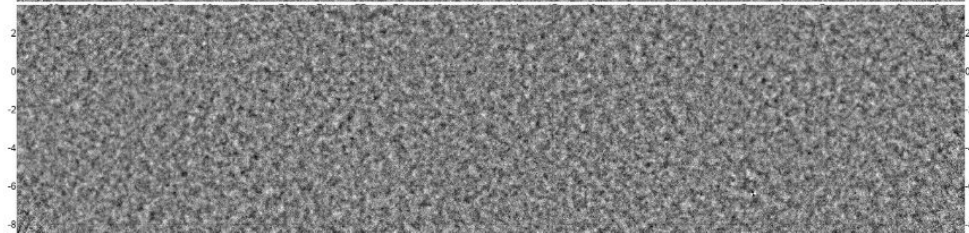
Q



U



E



ACT 2025

B

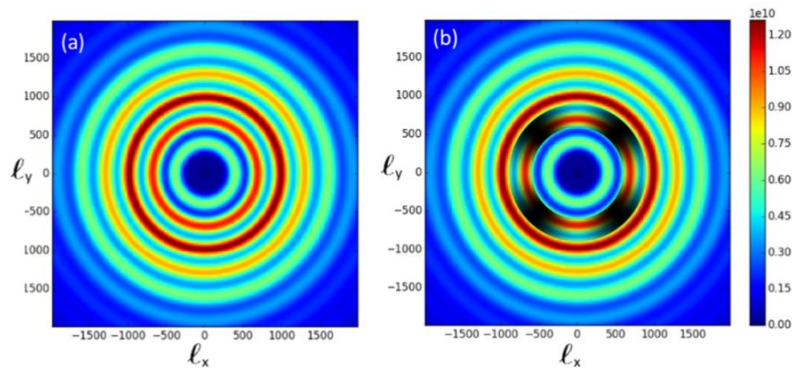


Why is there a plaid pattern?

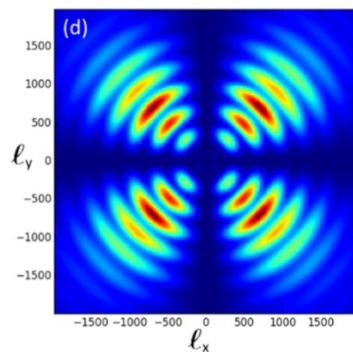
$$Q(\ell) = E(\ell) \cos(2\phi_\ell) - B(\ell) \sin(2\phi_\ell),$$

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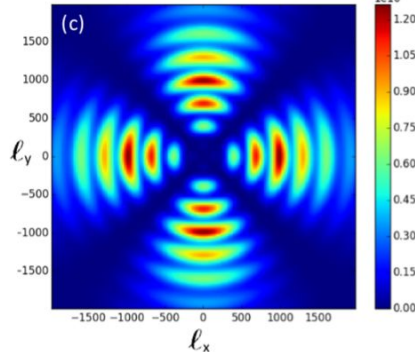
EE



UU



QQ



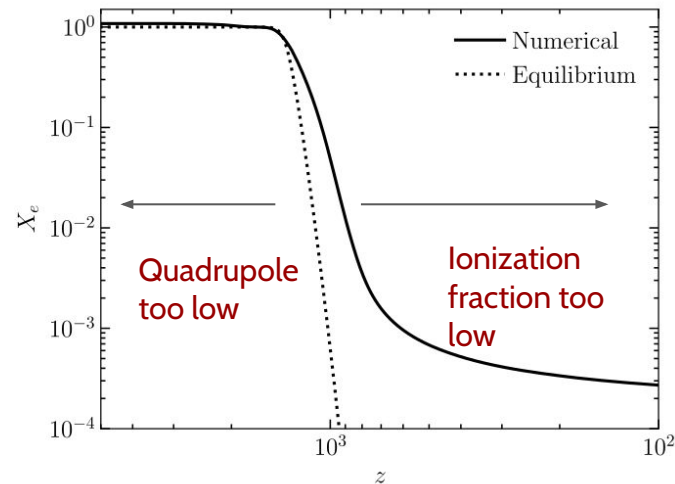
The CMB is E-polarized due to scalar perturbations:

- why is the polarization only ~10%?
- what acoustic peak structure do we get?

Pol needs both quadrupole and free electrons.

Polarization generated during brief period of time when quadrupole built up from free streaming at recombination, but with rapidly depleting fraction of free electrons.

Quadrupole generated from dipole -> measuring velocities



Links to CMB Analysis Resources

[A big list of notebooks and webinars](#)

- [CMB analysis from scratch](#)
- [pixell tutorial notebooks](#)
- [ACT DR6 notebooks](#)

Visualizing maps:

- [Tilemaker](#) to visualize maps
- Online hosted access to Tilemaker at maps.simonsobservatory.org
(Try searching for El Gordo)



Josh Borrow
Systems
Architect
(Penn)



Jeremy Myers
Front-end
engineer
(Penn)